

Geoarchaeology of Jefferson's Monticello Plantation

Fraser D. Neiman, Monticello

Sara Bob-Harper, Highland

Derek Wheeler, Monticello

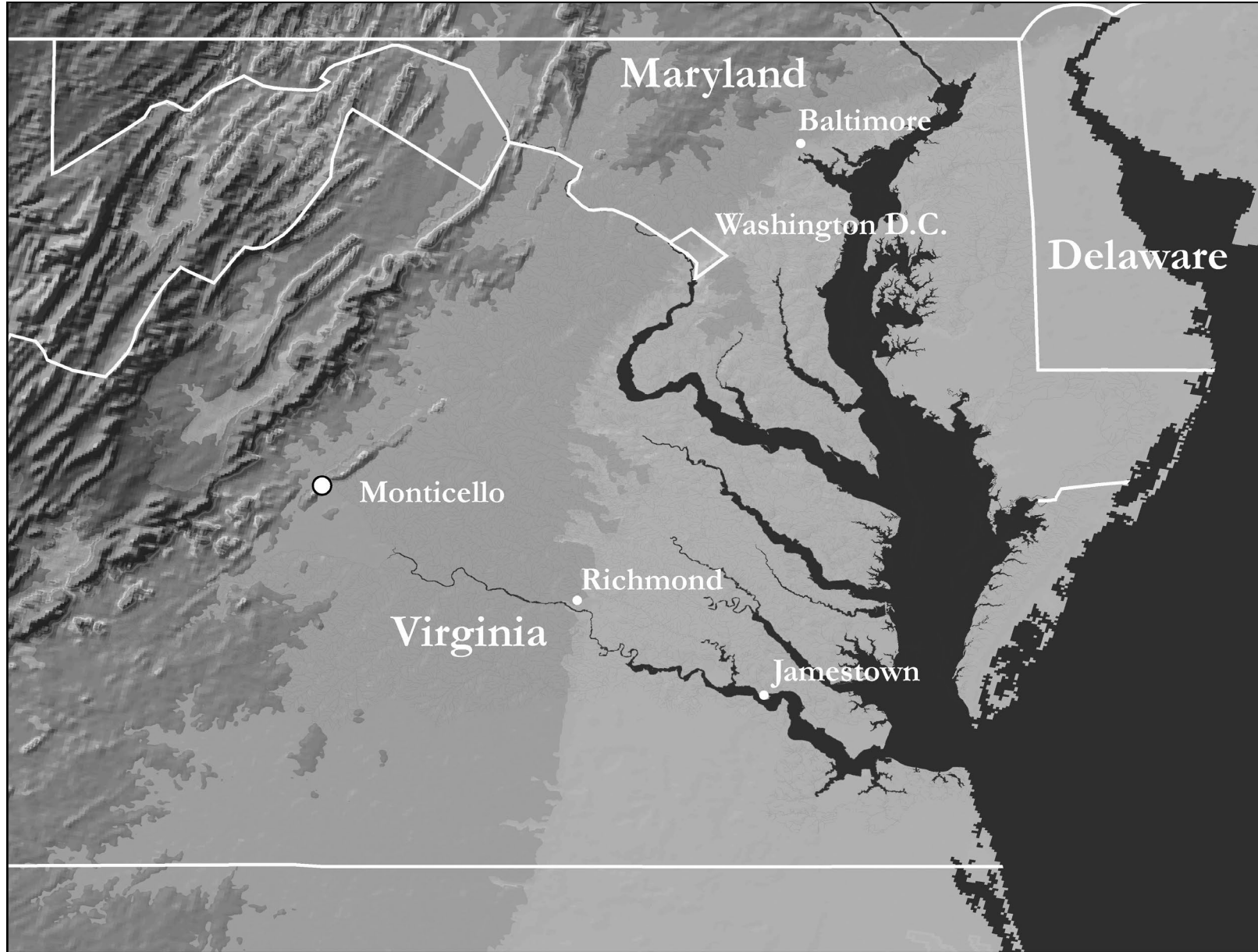
John G. Jones, Archaeological Consulting Services, Ltd.

James Burton, University of Wisconsin

James Feathers, University of Washington

<https://github.com/fneiman/MonticelloGeoArchy>

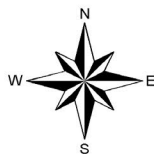




0 50 100 Miles

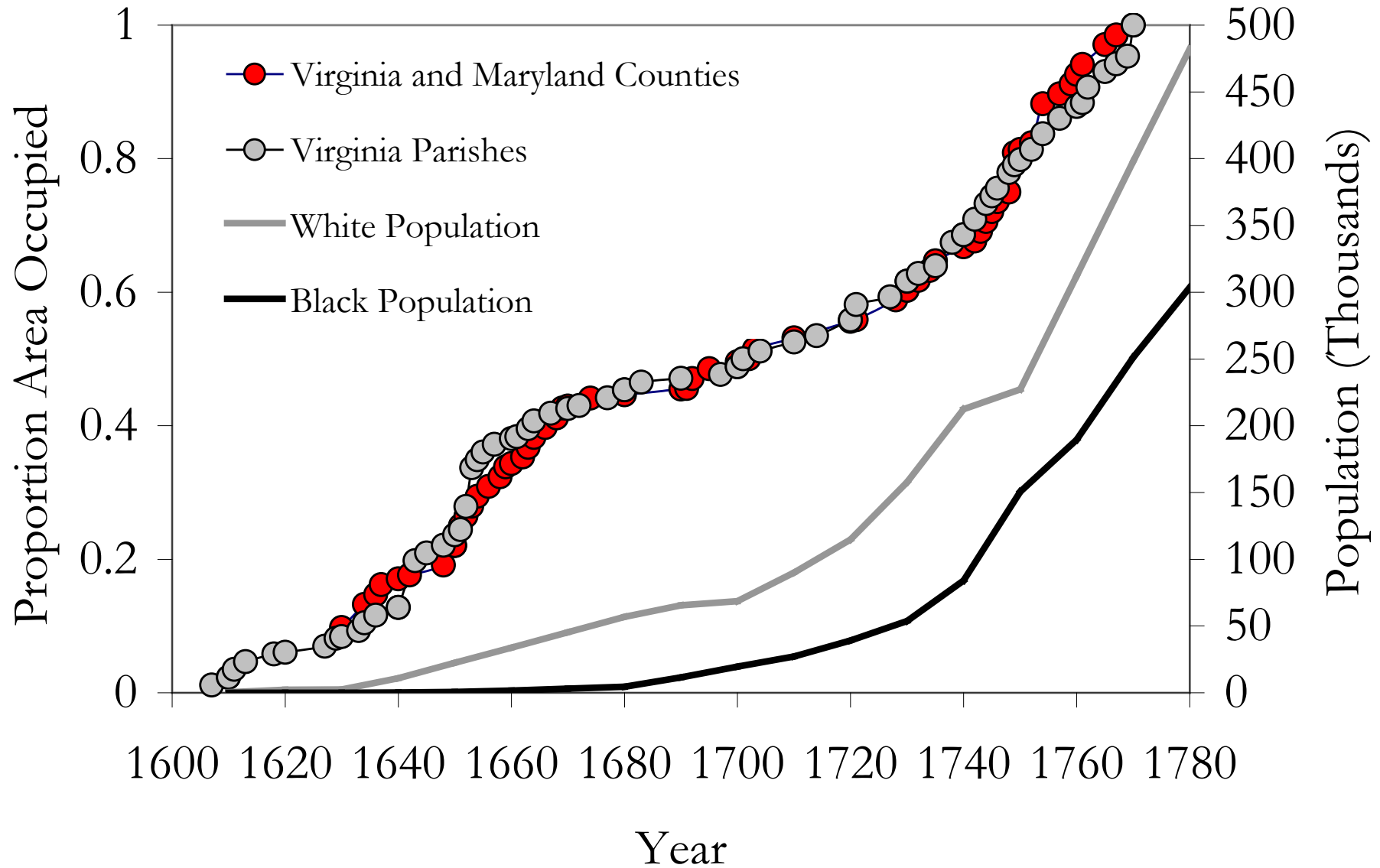


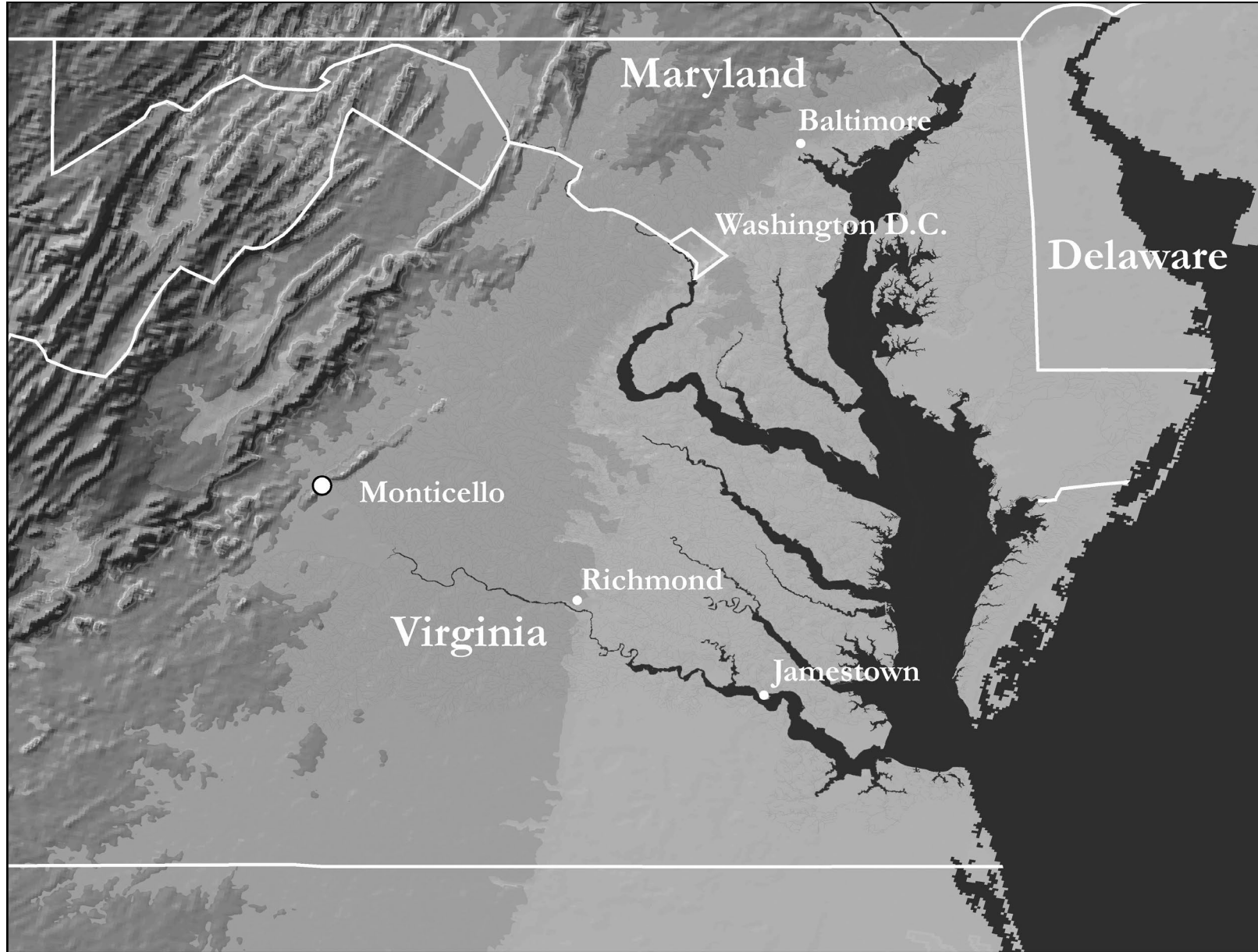
0 100 200 Kilometers





Demographic Expansion

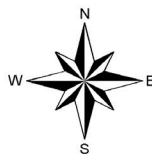


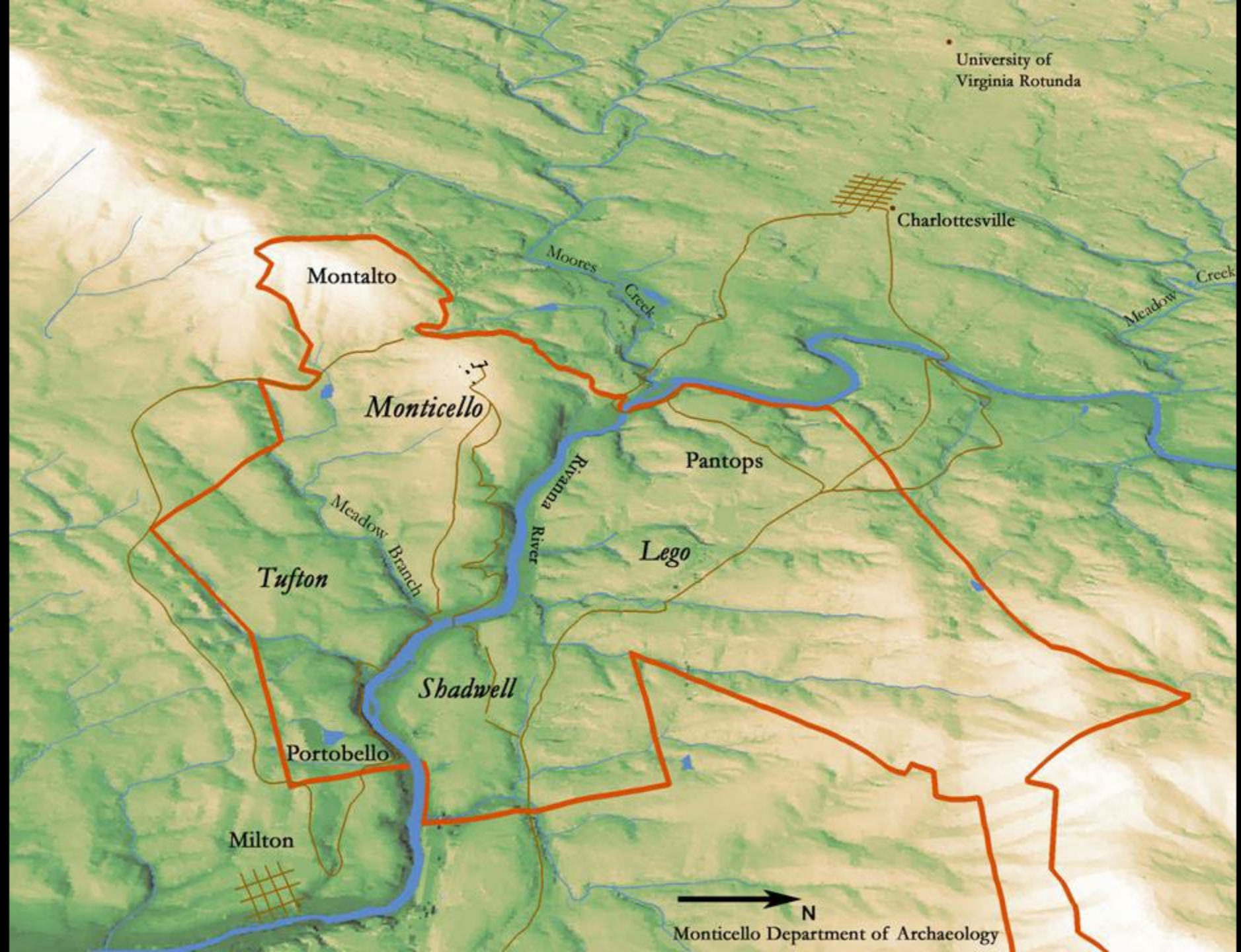


0 50 100 Miles

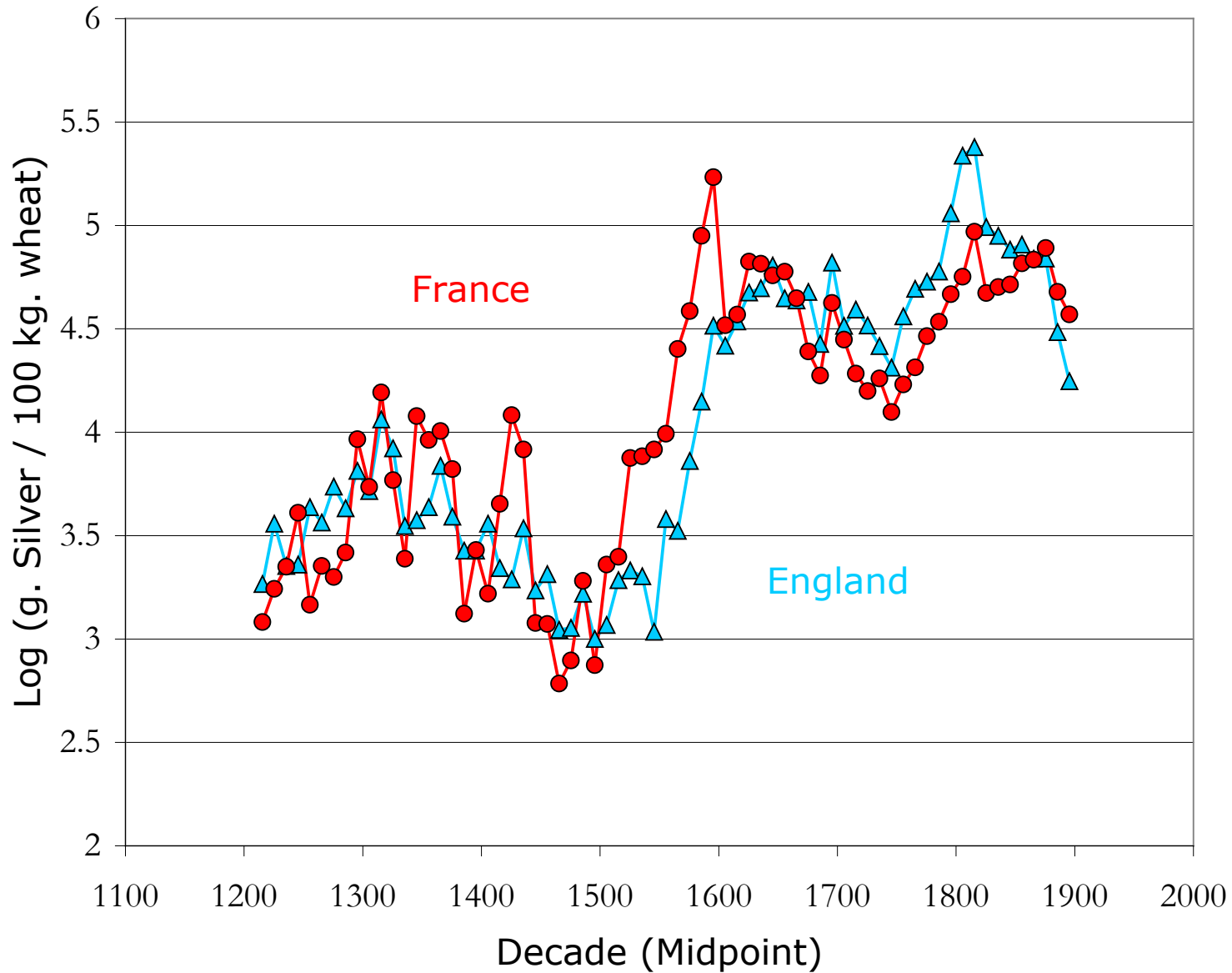


0 100 200 Kilometers





European Wheat Prices



*“There is reason to expect a rupture has taken place between Spain and England. If so, it will involve France and so render the present war of Europe almost universal there. I hope they will see it their interest to let us make bread for them in peace, and to give us a good price for it. **We have every moral certainty that wheat will be high for years to come. I cannot therefore, my dear Sir, omit to press, for myself, the going into that culture as much as you think practicable. In Albemarle I presume we may lay aside tobacco entirely; and in Bedford the more we can lay it aside, the happier I shall be.**”*

Thomas Jefferson to Nicholas Lewis
July 4, 1790



Tobacco

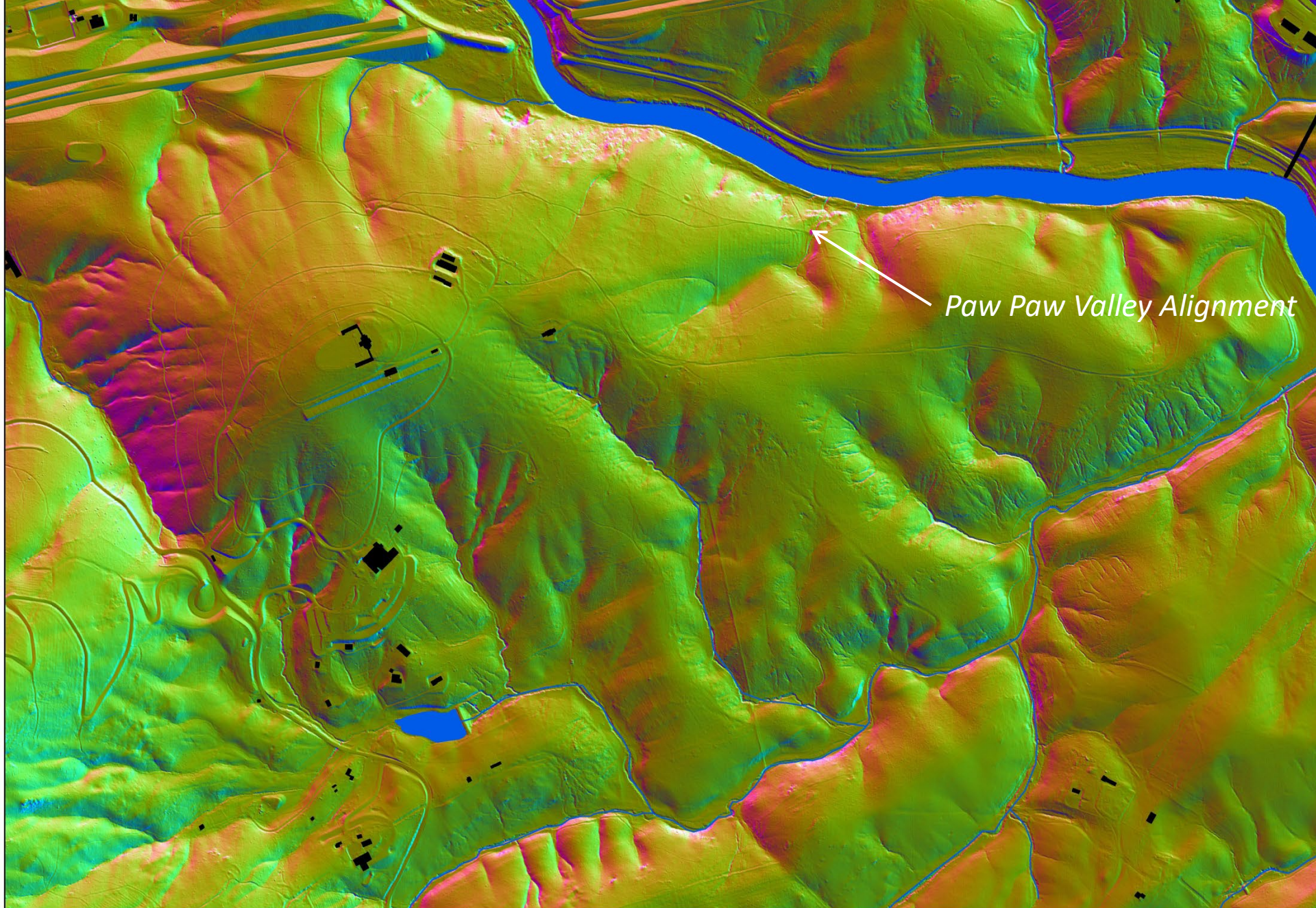
- Hoes
- Stumps
- Field rotation
- Long fallows for sustainability
- Locally evolved



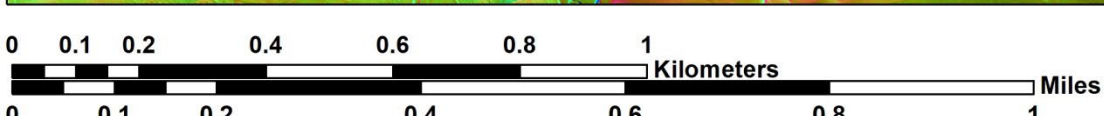
Wheat

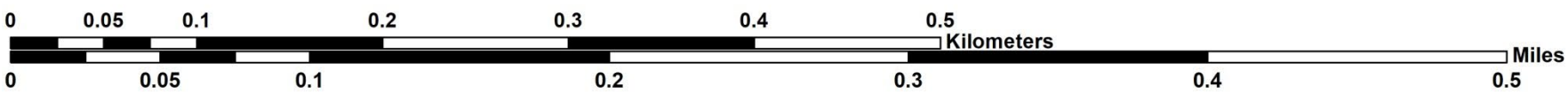
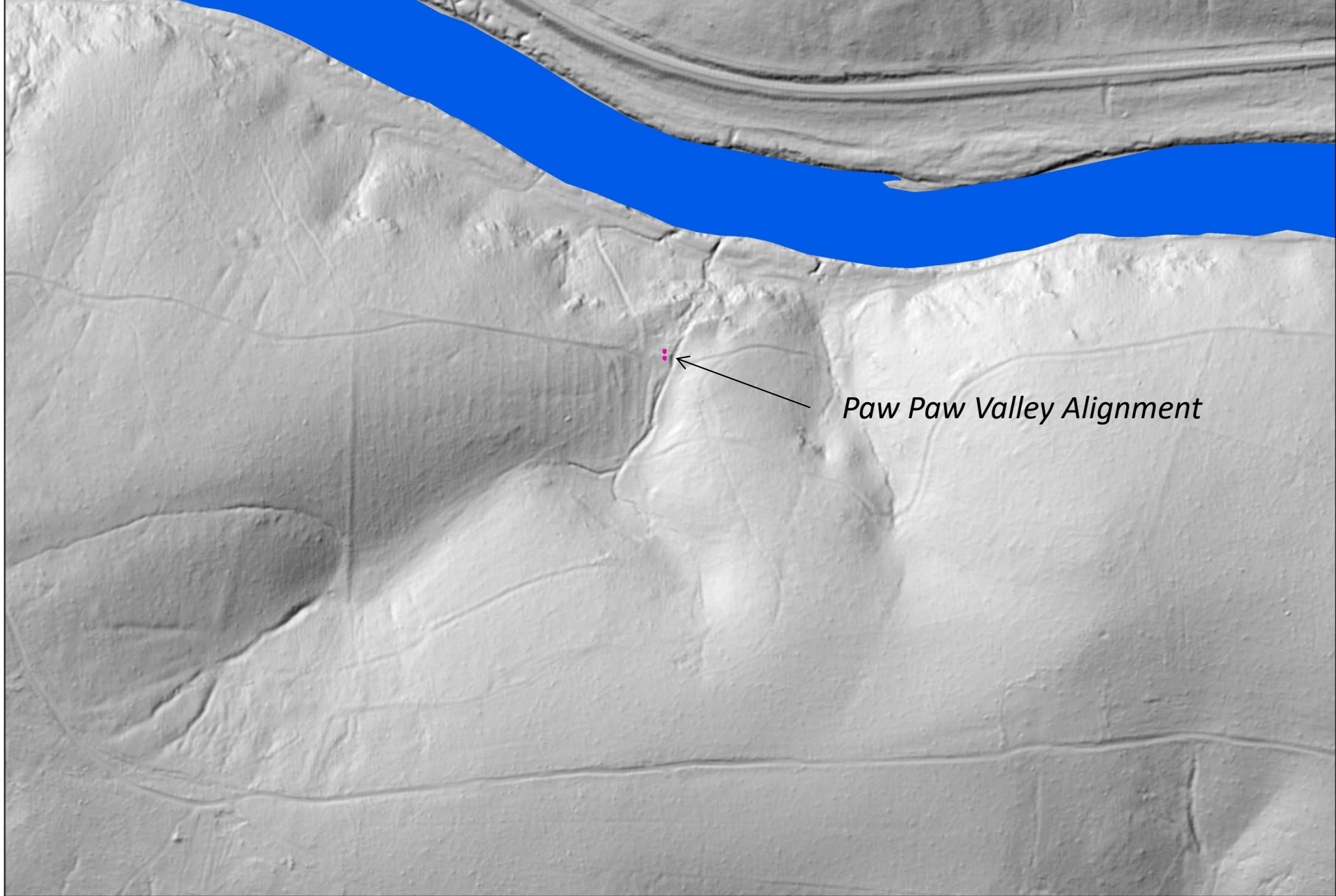
- Plows
- No stumps!
- Permanent fields
- Crop rotations and fertilizers for sustainability
- European handbooks





Paw Paw Valley Alignment





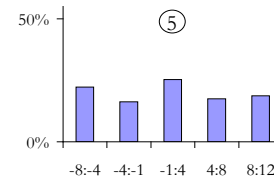
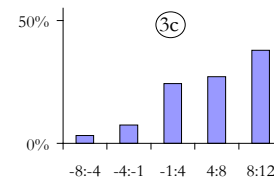
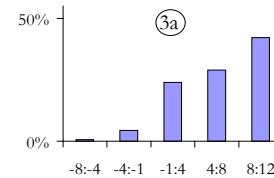
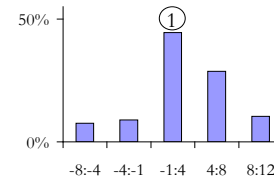
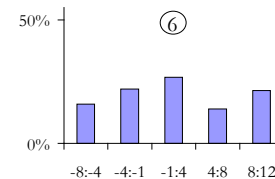
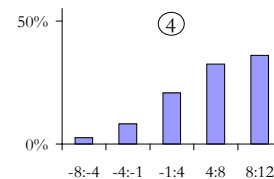
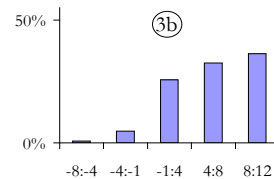
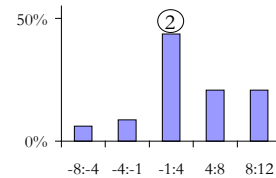
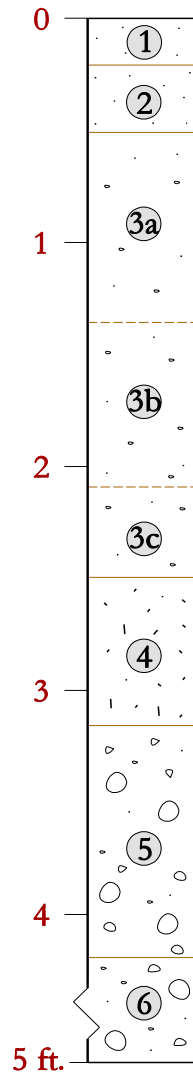




Paw Paw Valley: Unit 1

Lithostratigraphy and Grain Size

1. Reddish brown sandy loam.
2. Red sandy clay loam.
- 3a. Red clay with < 5% weathered greenstone pebbles.
- 3b. Red clay loam with < 5% weathered greenstone pebbles.
- 3c. Red clay with < 5% weathered greenstone pebbles.
4. Reddish brown clay with < 5% pebble-sized charcoal flecks.
5. Yellowish red clay loam with 25% pebbles and cobbles.
6. Strong brown clay loam with 10% weathered greenstone gravel and cobbles

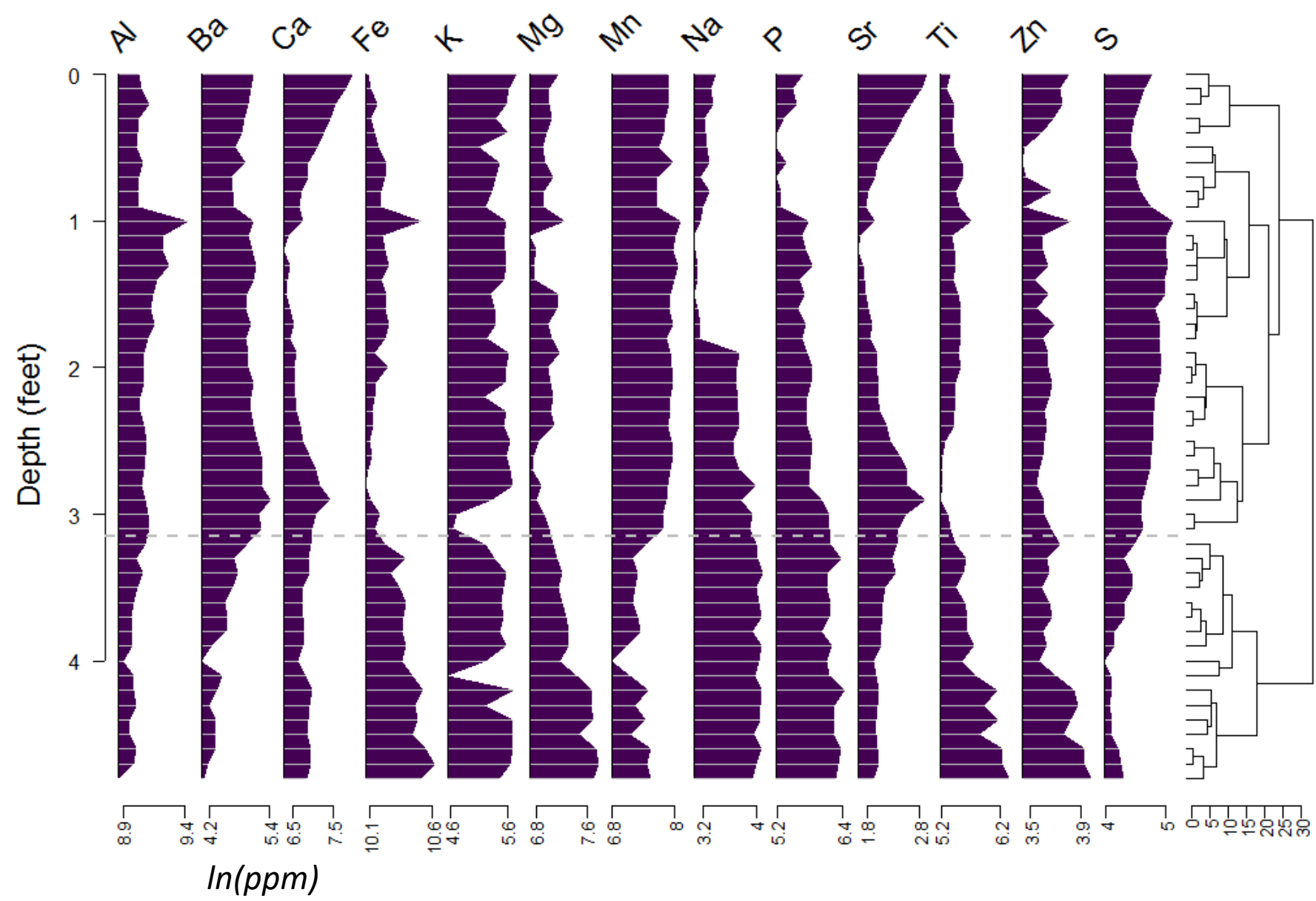


Cobbles-Pebbles-Sand-Silt-Clay

Phi

Paw Paw Valley: Unit 1

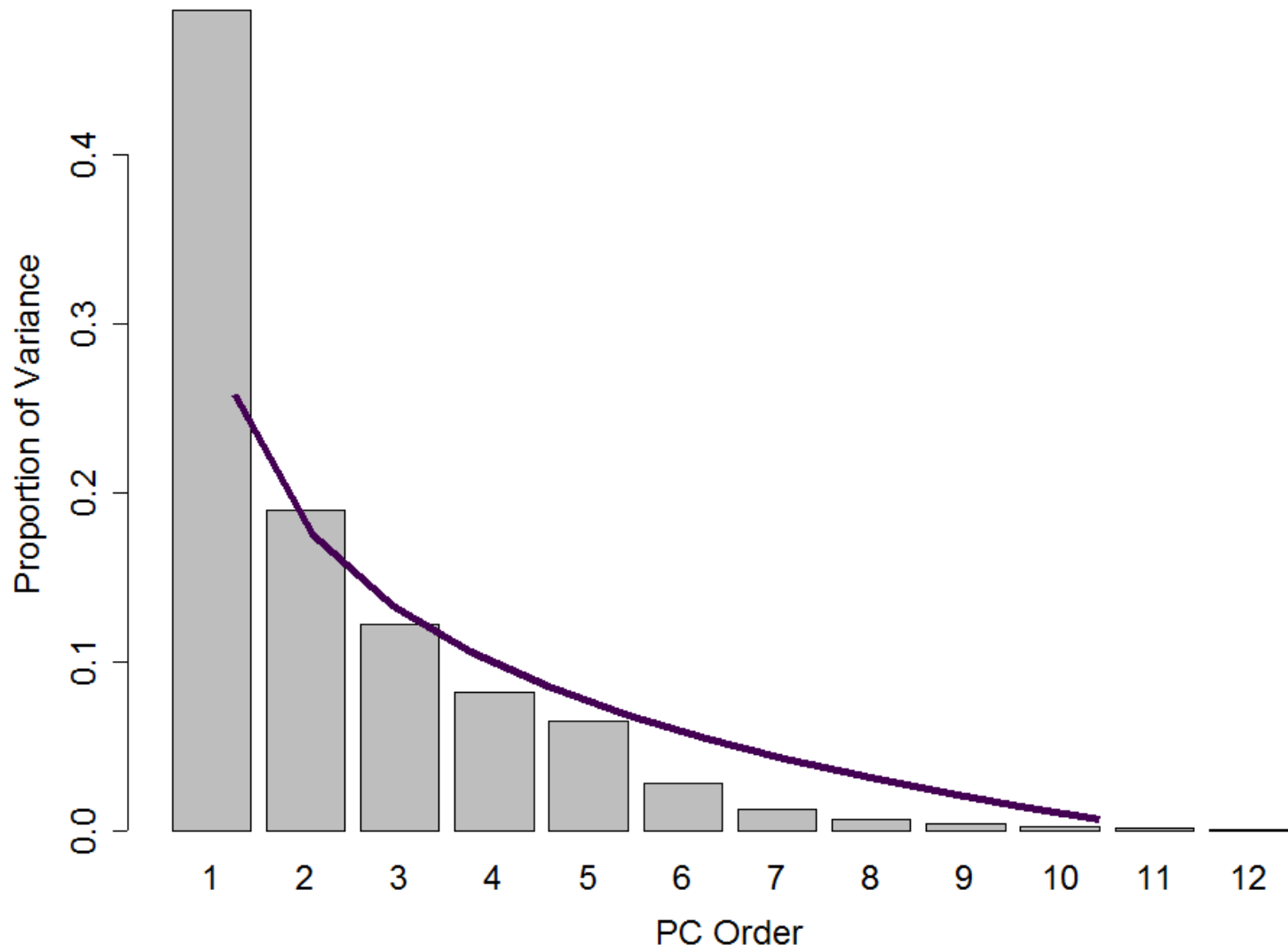
Sediment Chemistry



Paw Paw Valley: Unit 1

Sediment Chemistry

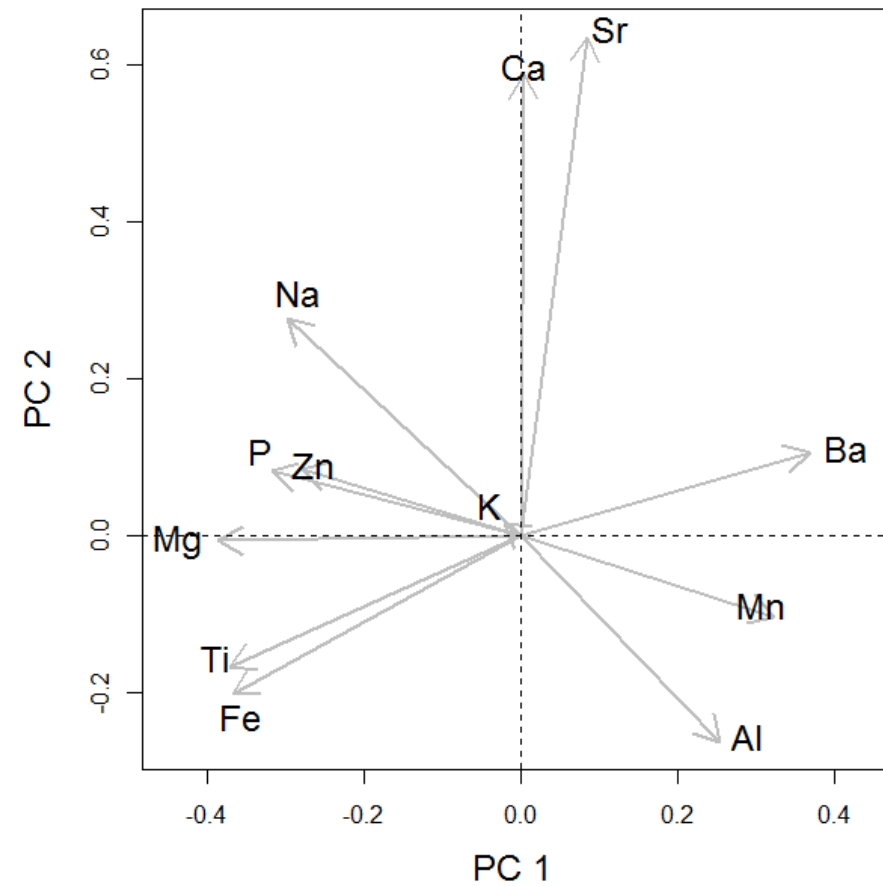
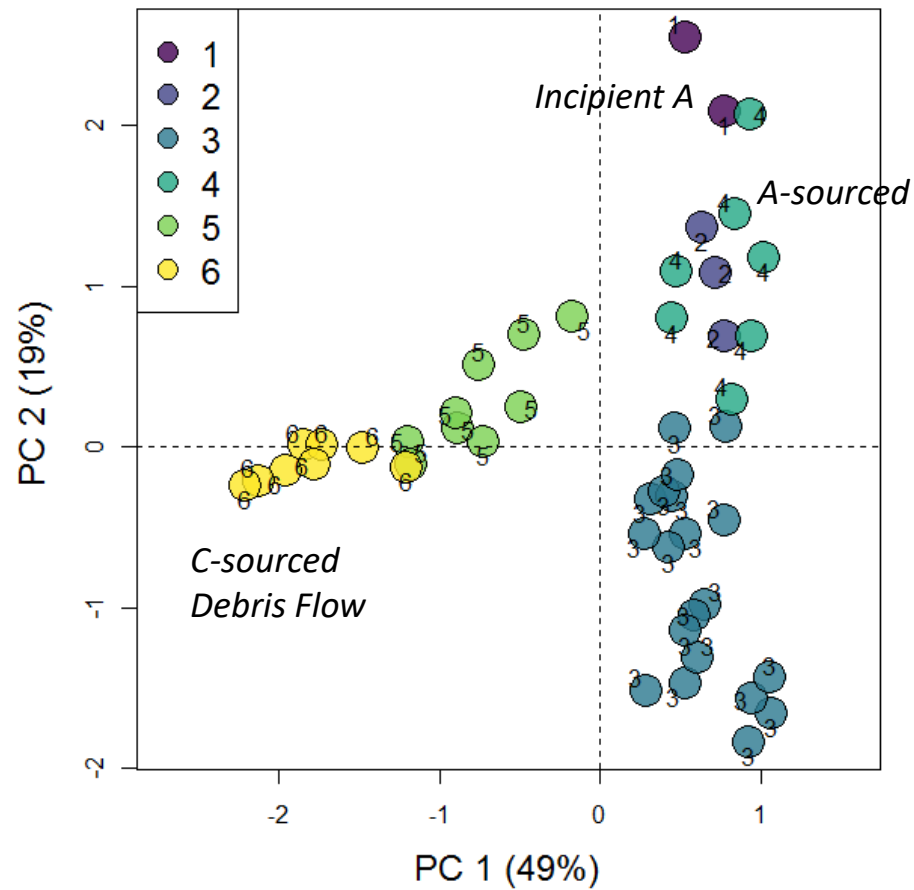
PCA



Paw Paw Valley: Unit 1

Sediment Chemistry

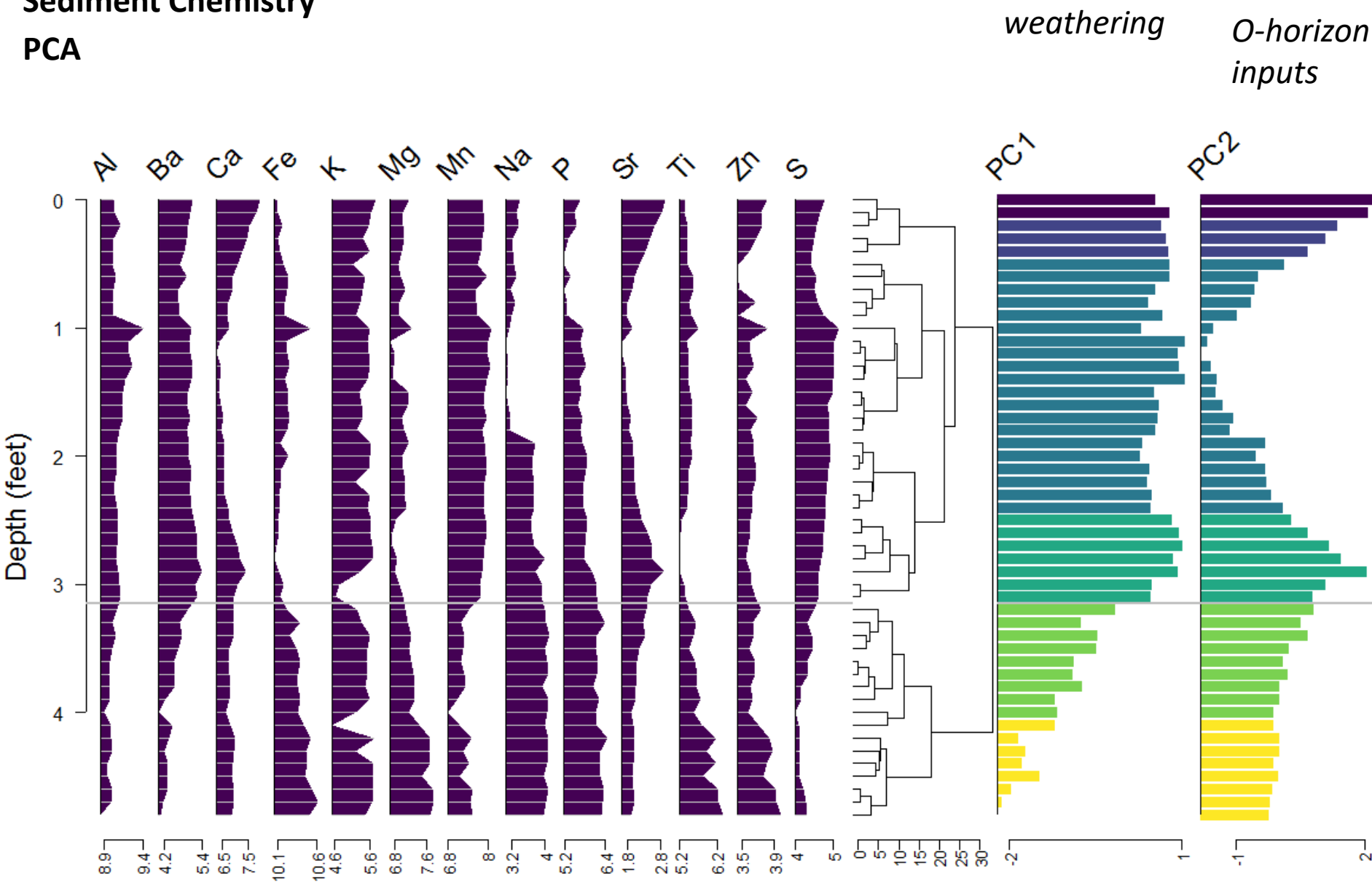
PCA



Paw Paw Valley: Unit 1

Sediment Chemistry

PCA



Paw Paw Valley: Unit 1

Lithostratigraphy and Grain Size

1. Reddish brown sandy loam.

2. Red sandy clay loam.

3a. Red clay with < 5% weathered greenstone pebbles.

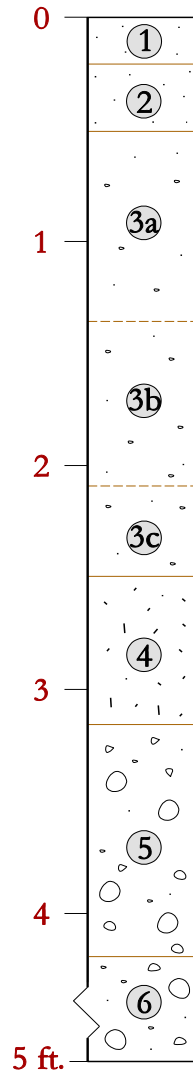
3b. Red clay loam with < 5% weathered greenstone pebbles.

3c. Red clay with < 5% weathered greenstone pebbles.

4. Reddish brown clay with < 5% pebble-sized charcoal flecks.

5. Yellowish red clay loam with 25% pebbles and cobbles.

6. Strong brown clay loam with 10% weathered greenstone gravel and cobbles



1-2.

- Colluvium reaches the top of the wall.
- Incipient A horizon development
- Illuviation of O-horizon inputs
- Sand deposited from decelerating runoff

3.

- B horizon transported from upslope

4.

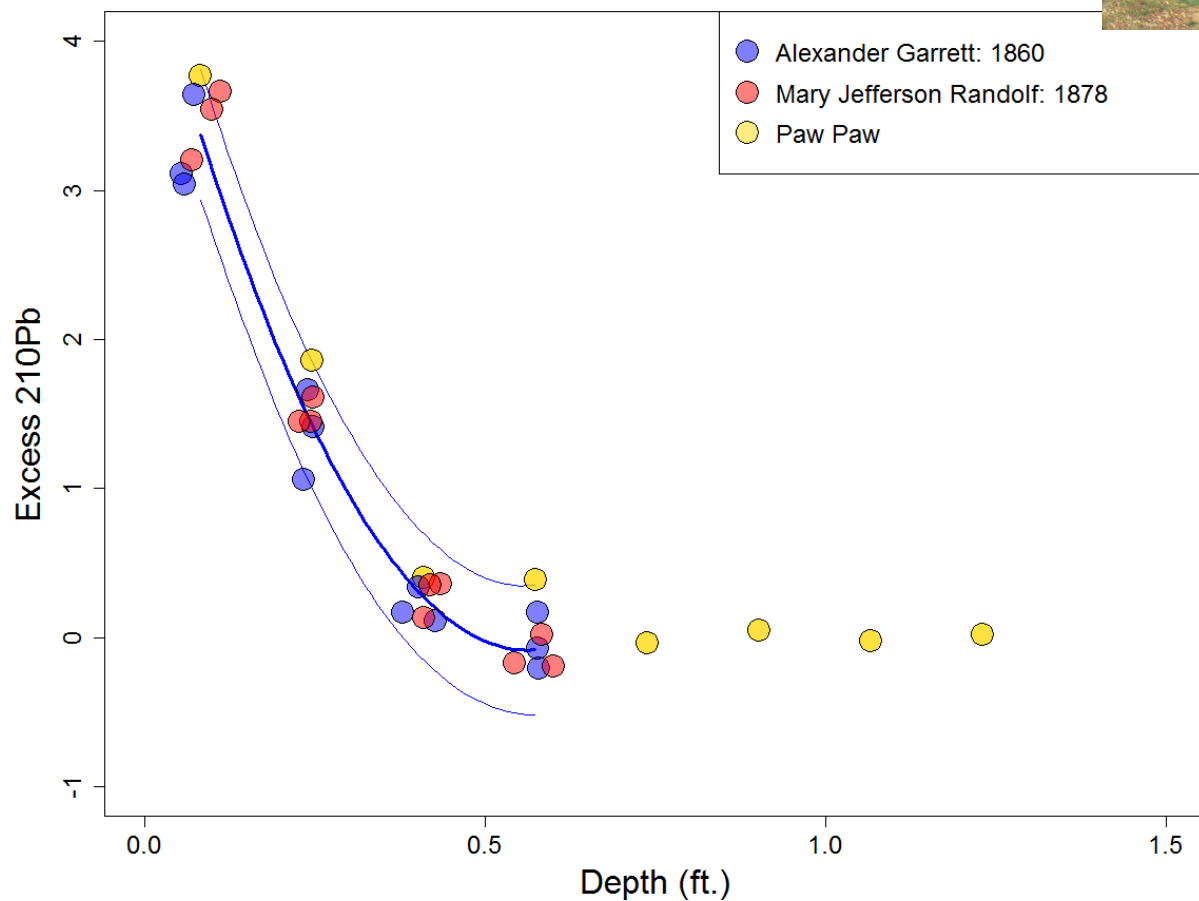
- A horizon **transported** from upslope

5-6.

- Massive debris flow.
- Weathering gradient.
- Stream bed, hence missing A-horizon.
- Some Illuviation of O-horizon inputs from stream flow.

Paw Paw Valley: Unit 1

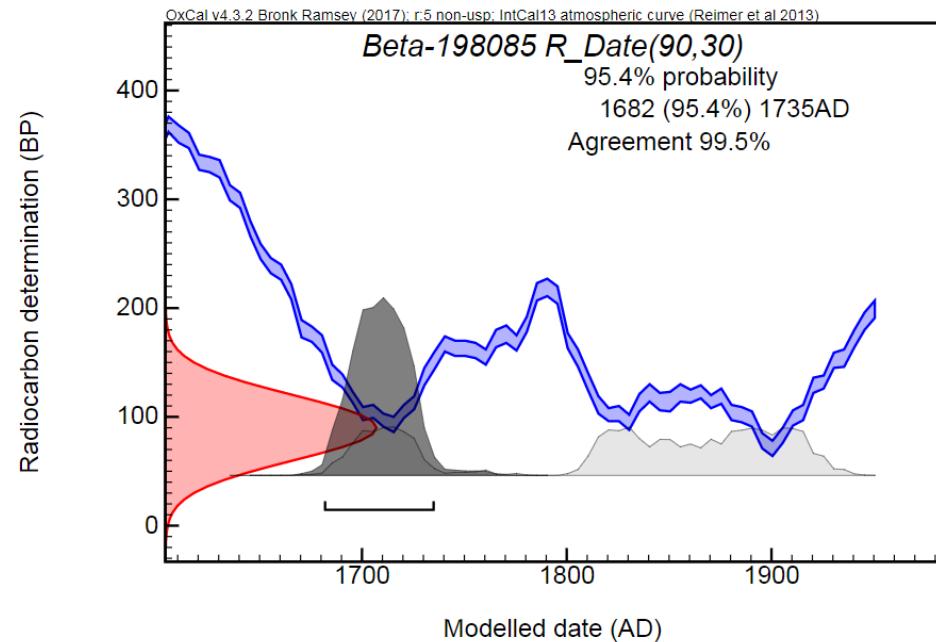
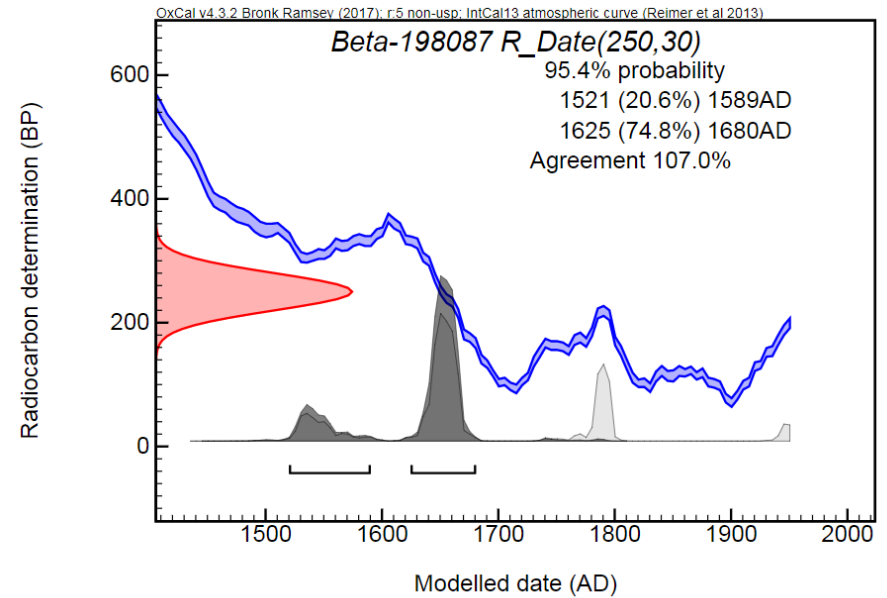
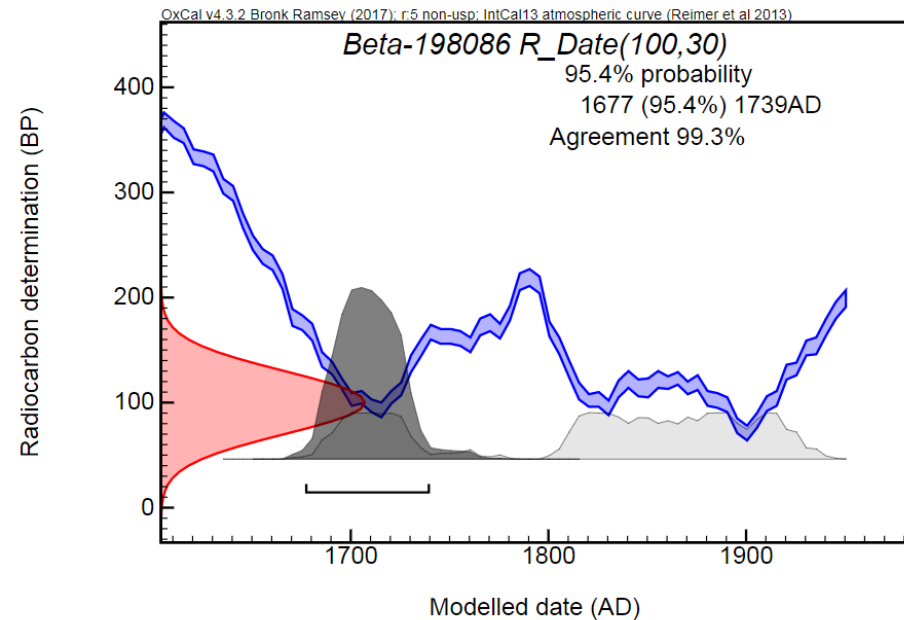
Dating: ^{210}Pb



< 1900

Paw Paw Valley: Unit 1

Dating: 14C



Paw Paw Valley: Unit 1

Bayesian Chronological Model

1. We have some good data on dates, but with some uncertainty – the **likelihood** .
2. We also have some good **prior** information about what date are likely. *E.g.* from stratigraphic relationships among the dates

We want away to combine 1 and 2 to come up with...

3. More accurate estimates of dates with LESS uncertainty – the “**posterior estimate**”.

Posterior

*Probability that event
E happened in year Y,
given our data.*

$\propto$

Likelihood

*Probability of getting
our data, given that
event E happened in
year Y.*

$\times$

Prior

*Prior probability that
event E happened in
year Y.*

Paw Paw Valley: Unit 1

Posterior Estimates from the Bayesian Chronological Model

Mean (95% credible interval)

14C: 1636 (1530 – 1742)

14C: 1706 (1680 – 1731)

14C: 1706 (1681 – 1730)

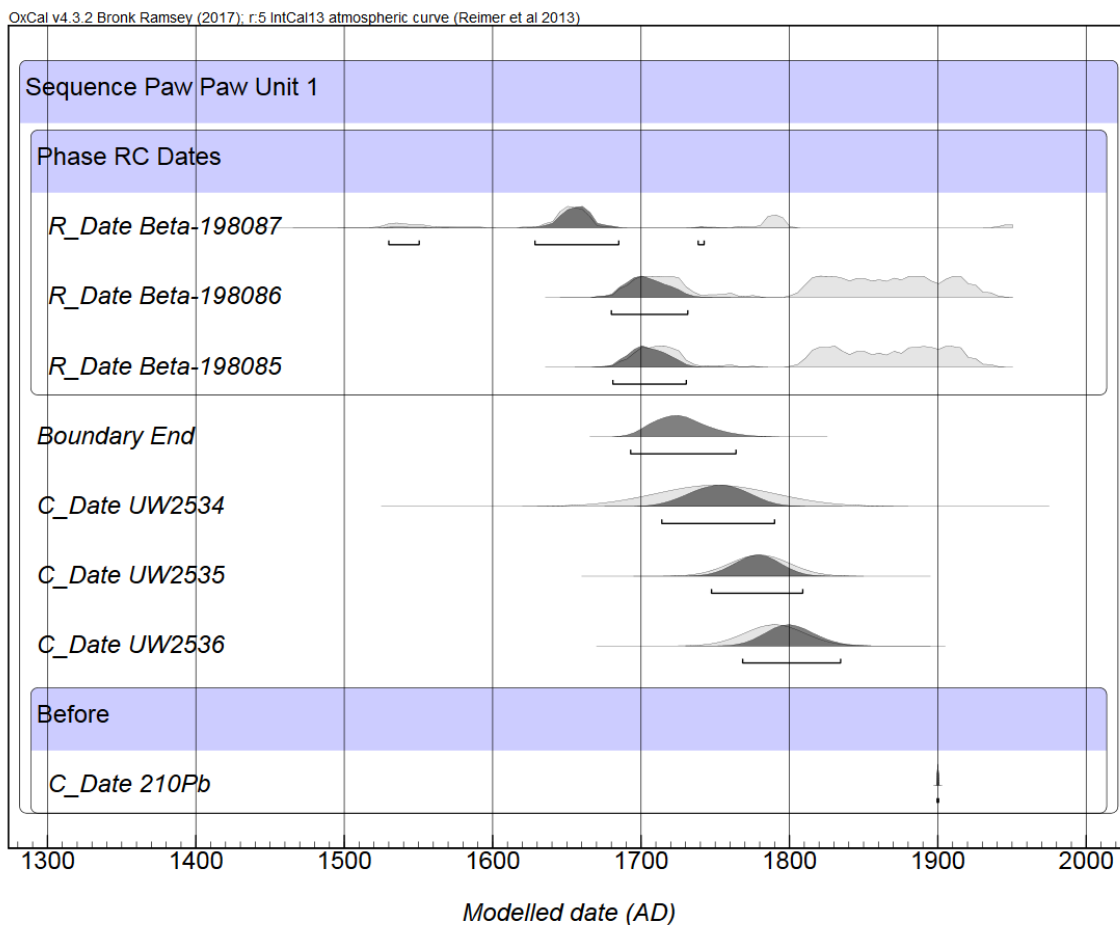
1728 (1693 – 1764)

OSL: 1752 (1714 – 1790)

OSL: 1778 (1747 – 1809)

OSL: 1801 (1768 – 1834)

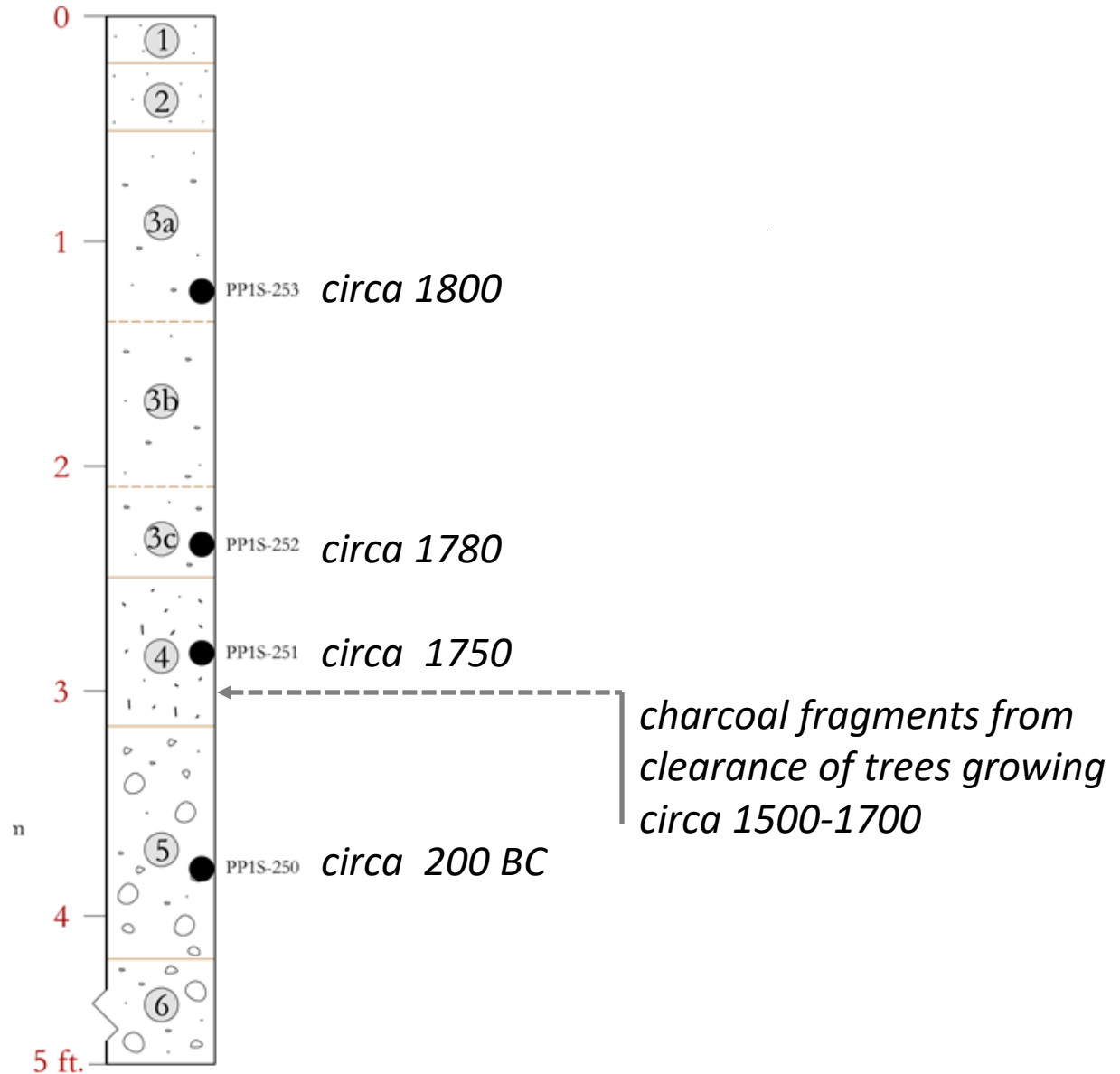
210Pb: < 1900



Paw Paw Valley: Unit 1

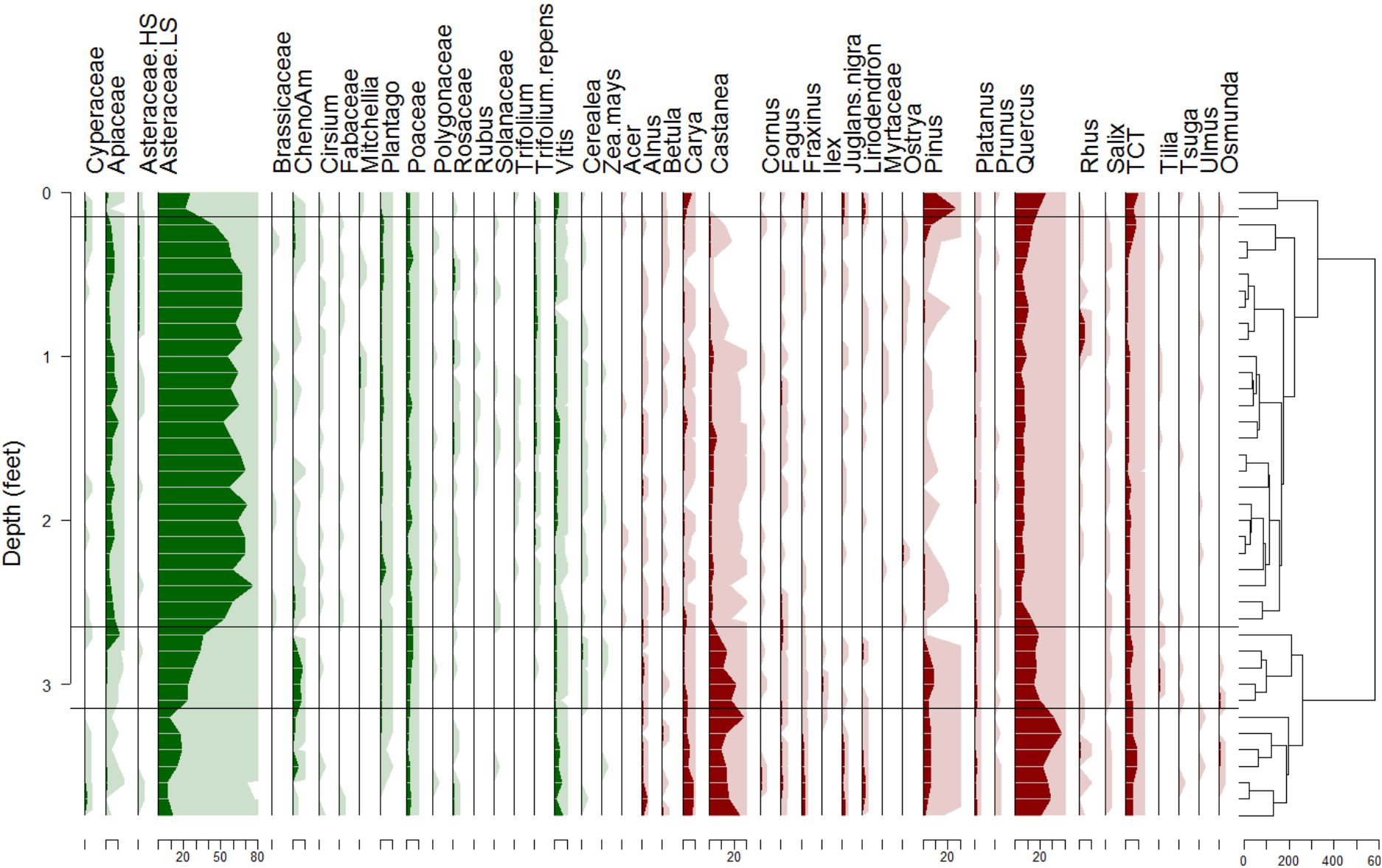
Proposed Chronology

1. Reddish brown sandy loam.
2. Red sandy clay loam.
- 3a. Red clay with < 5% weathered greenstone pebbles.
- 3b. Red clay loam with < 5% weathered greenstone pebbles.
- 3c. Red clay with < 5% weathered greenstone pebbles.
4. Reddish brown clay with < 5% pebble-sized charcoal flecks.
5. Yellowish red clay loam with 25% pebbles and cobbles.
6. Strong brown clay loam with 10% weathered greenstone gravel and cobbles



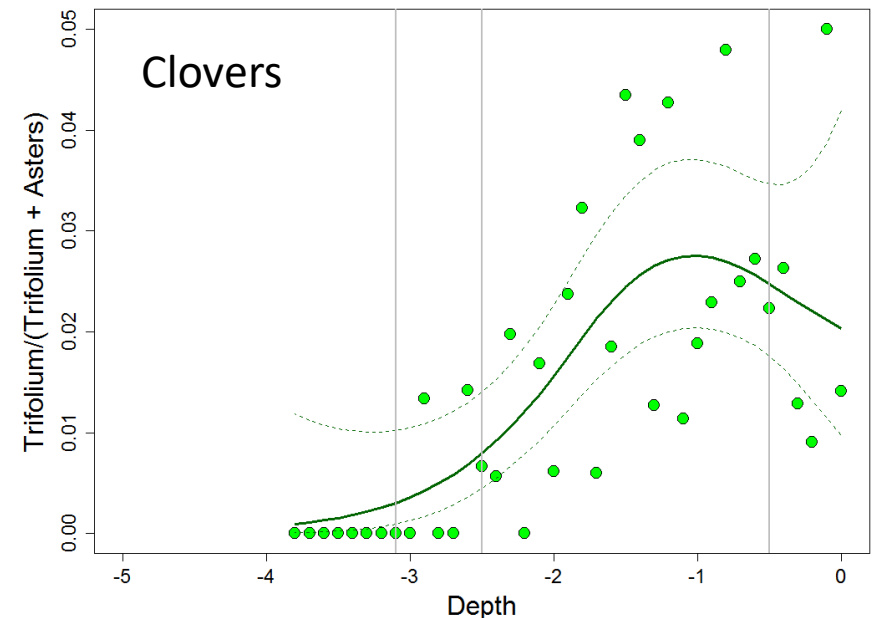
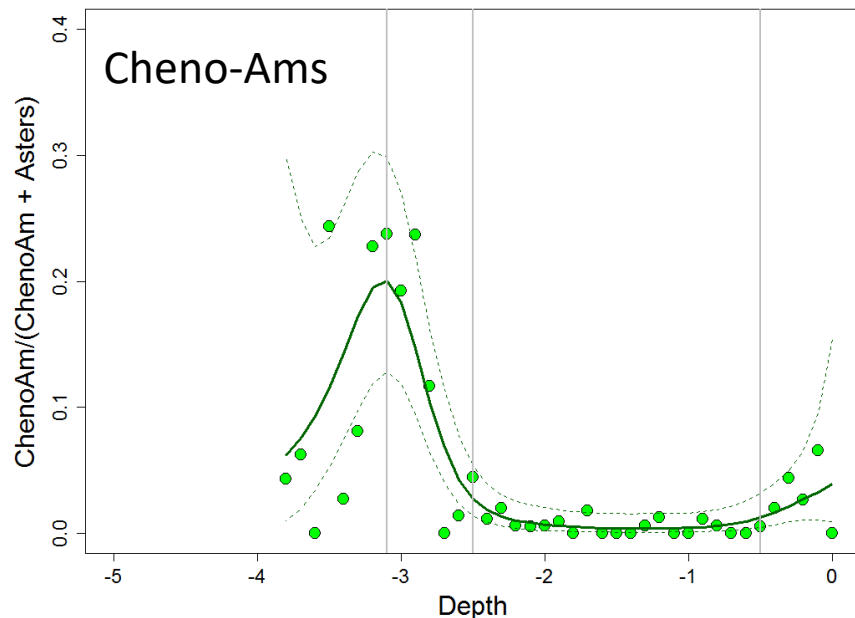
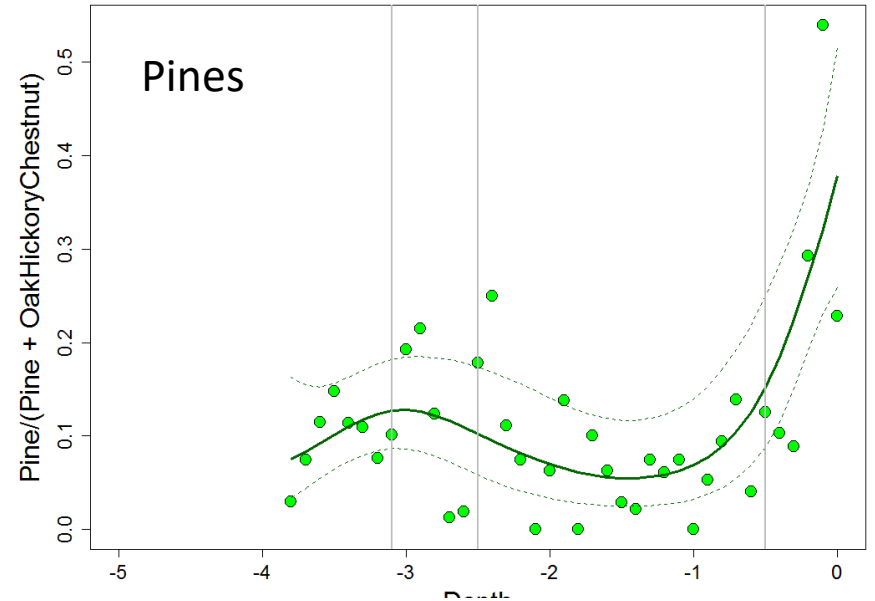
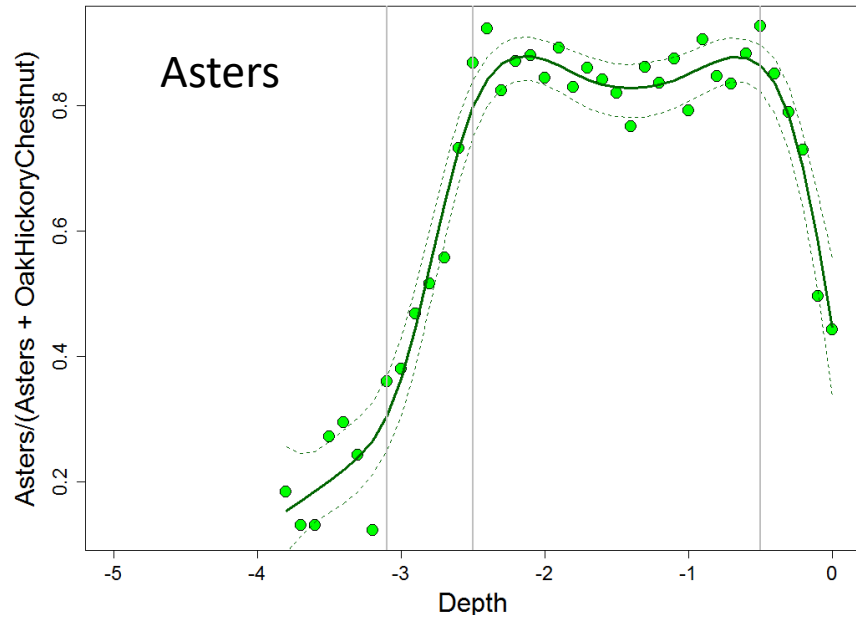
Paw Paw Valley: Unit 1

Pollen



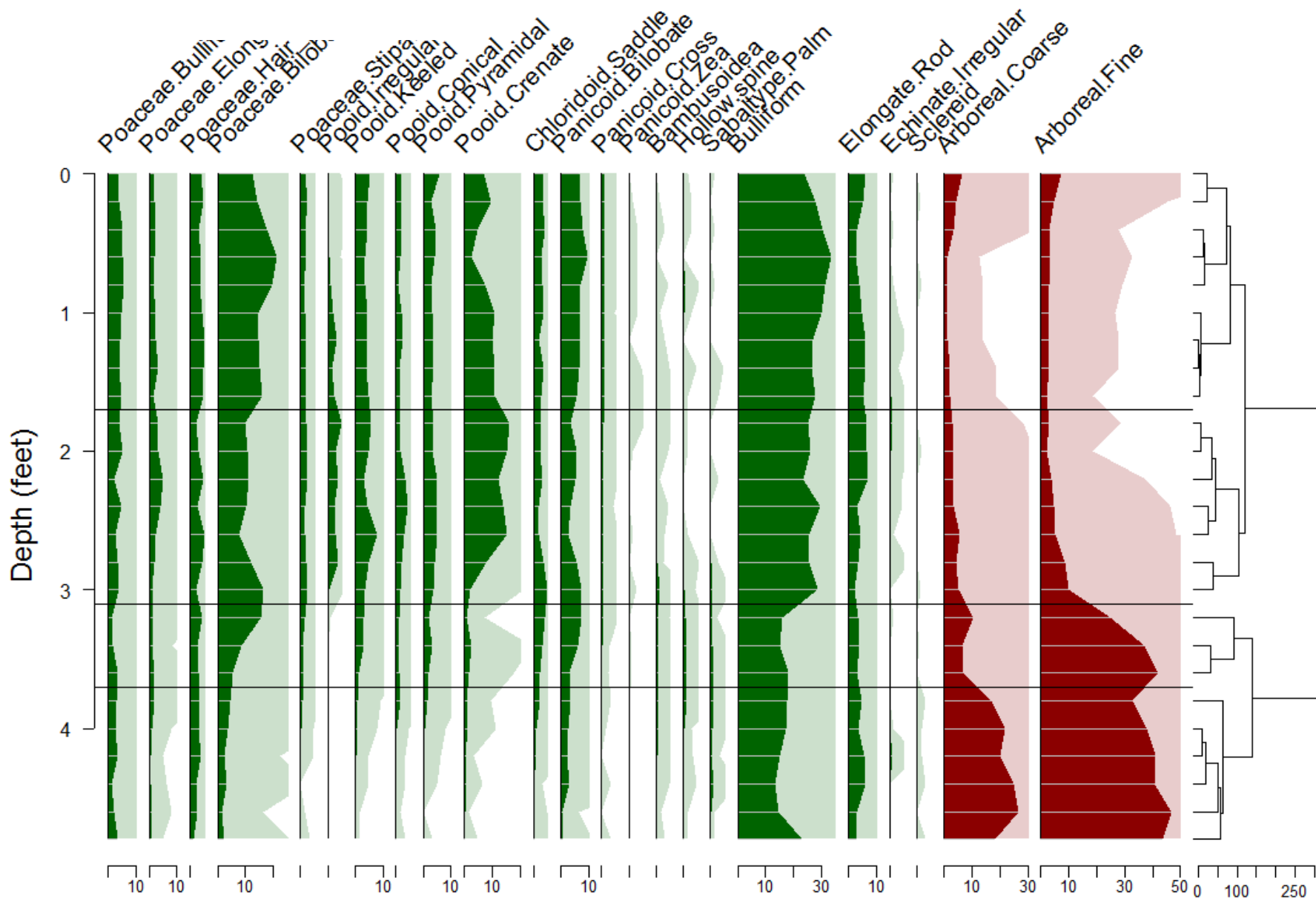
Paw Paw Valley: Unit 1

GAMs for Select Pollen Taxa: Logistic link, quasi-binomial errors



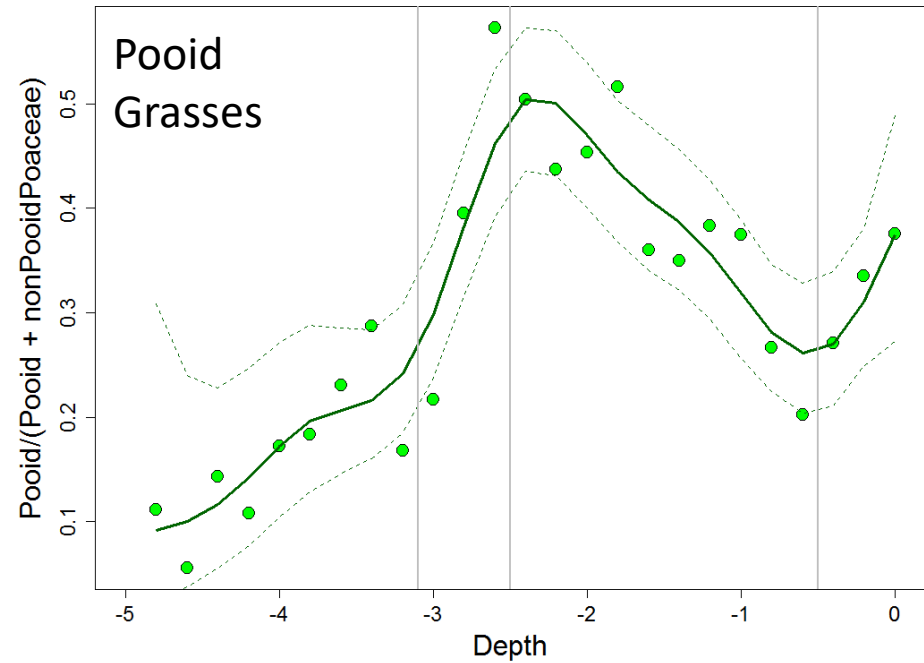
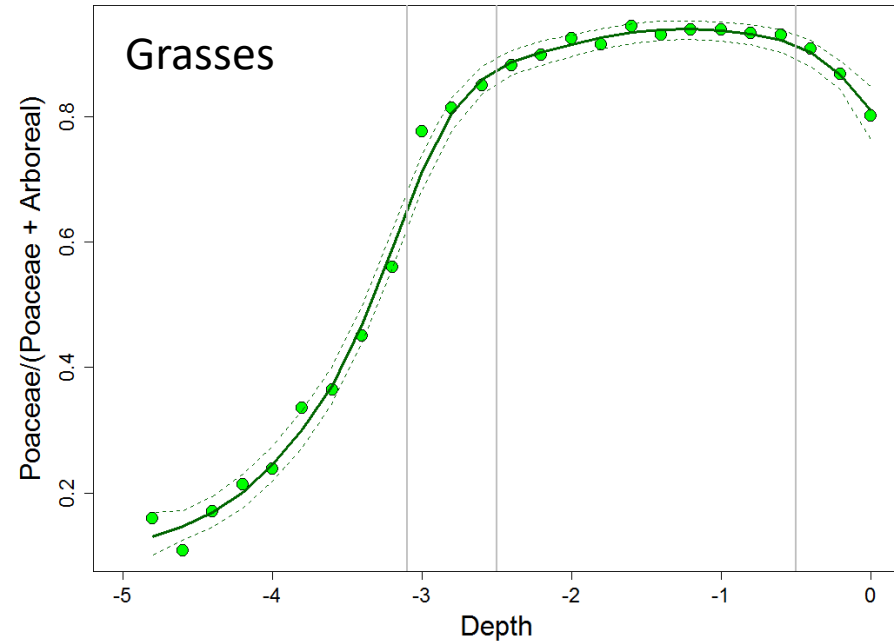
Paw Paw Valley: Unit 1

Phytoliths



Paw Paw Valley: Unit 1

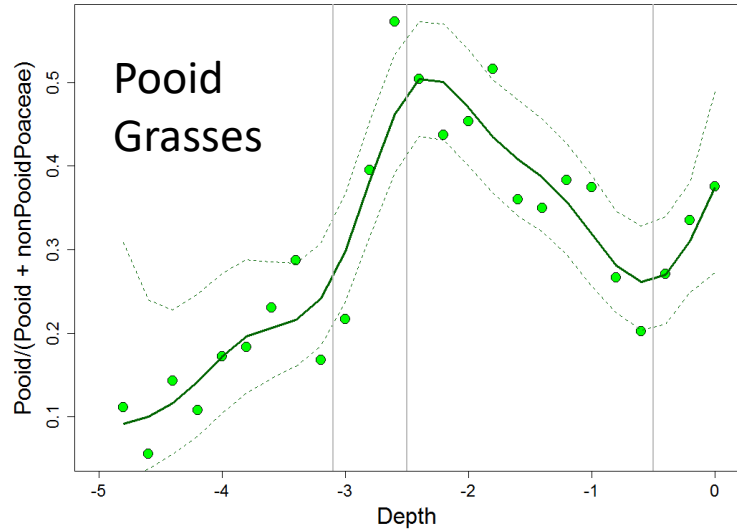
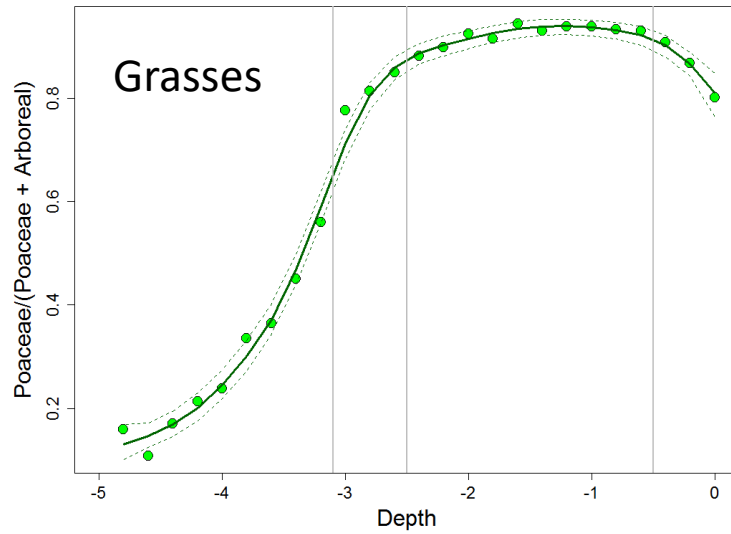
GAMs for Select Phytolith Taxa: Logistic link, quasi-binomial errors



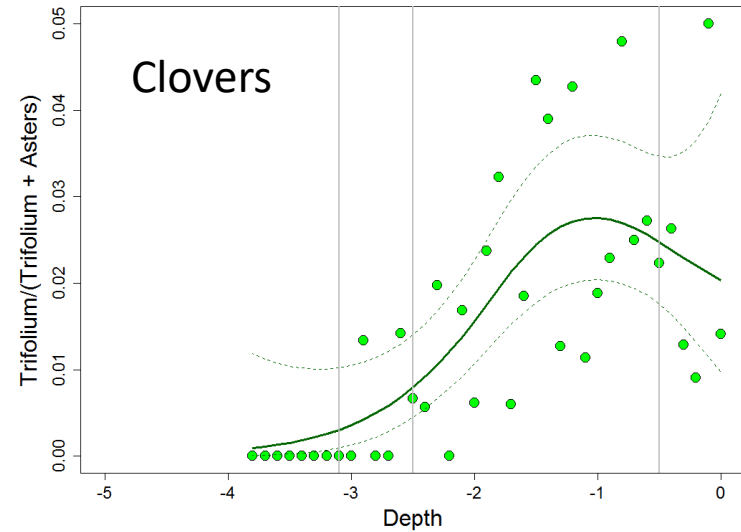
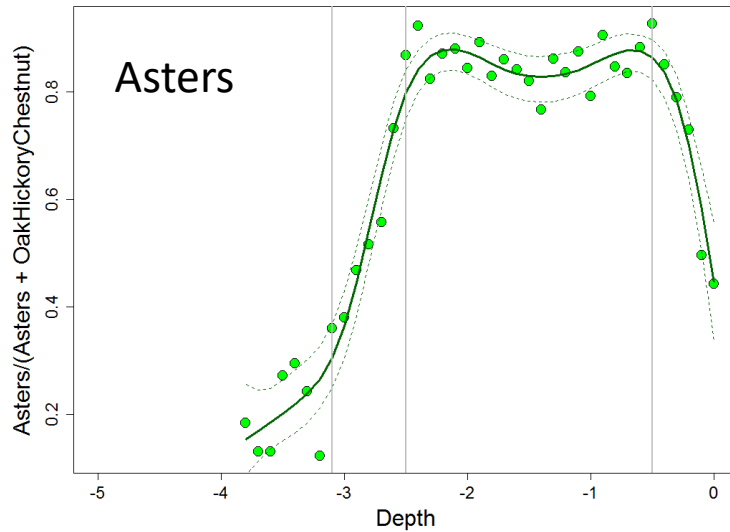
Paw Paw Valley: Unit 1

GAMs for Phytolith *versus* Pollen Taxa

Phytoliths

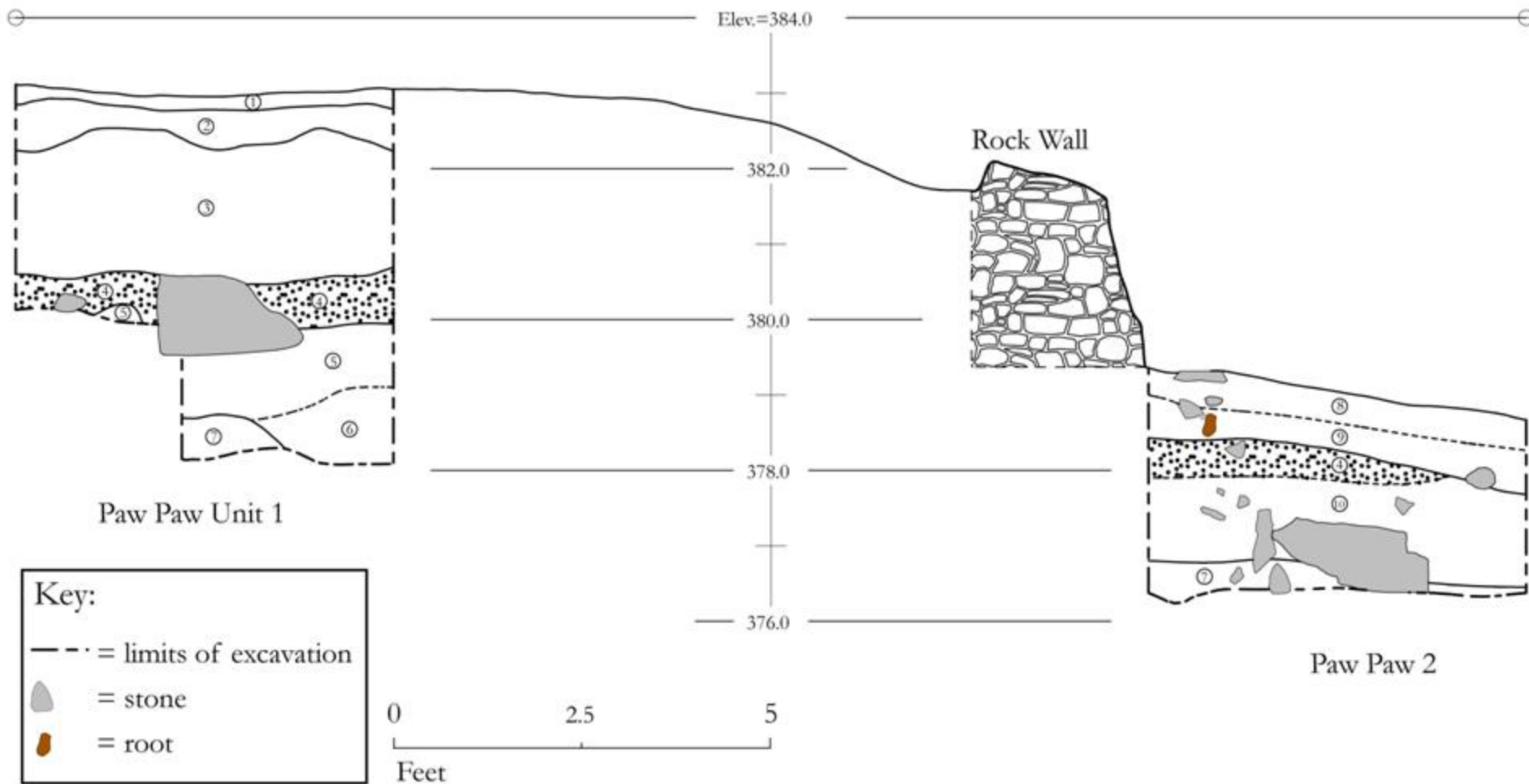


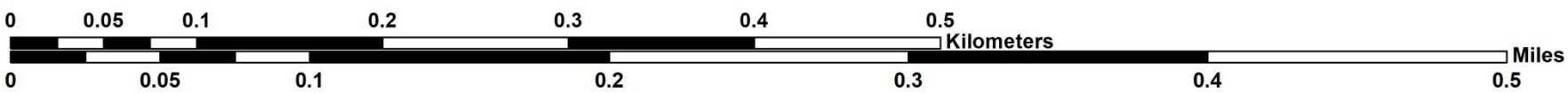
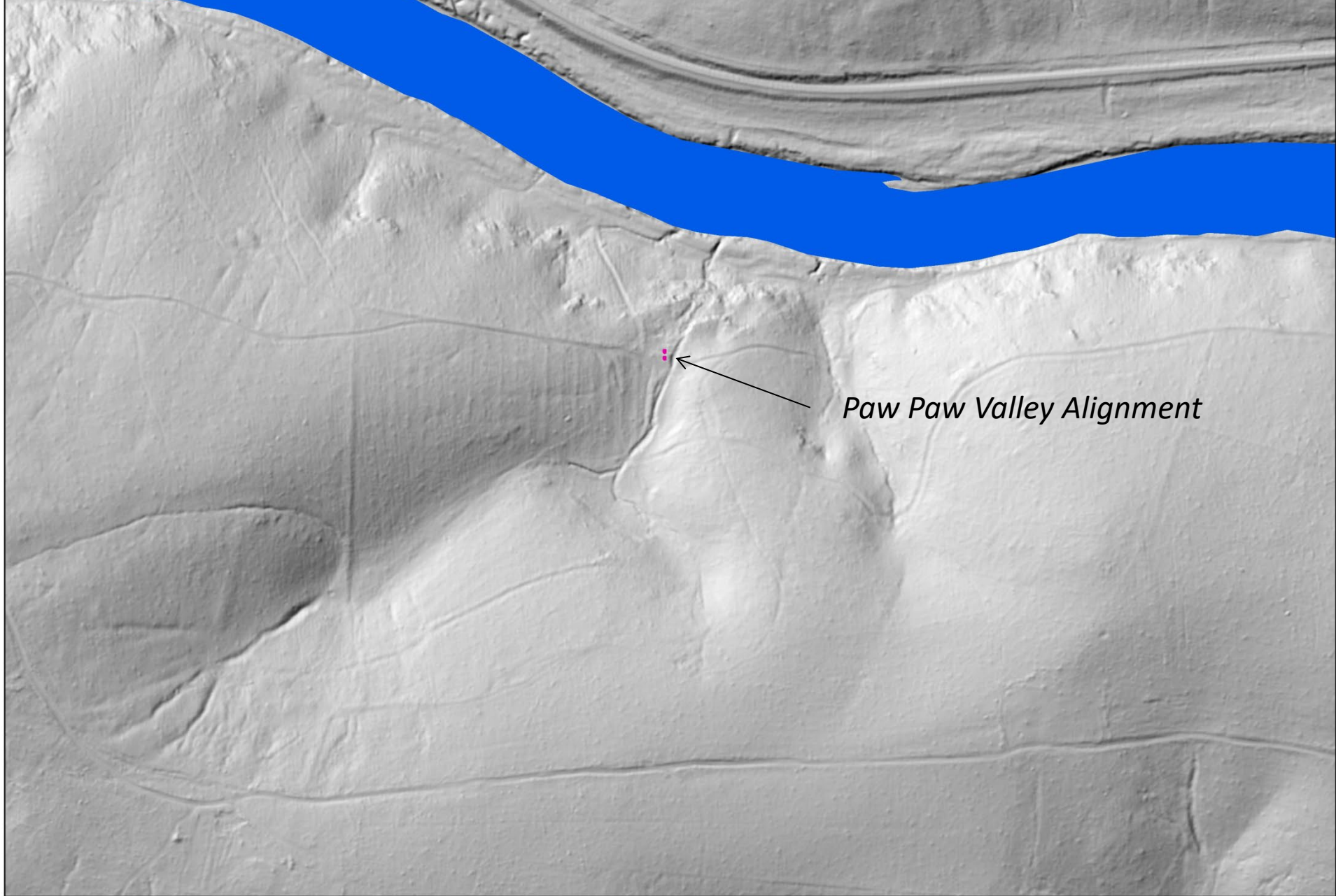
Pollen

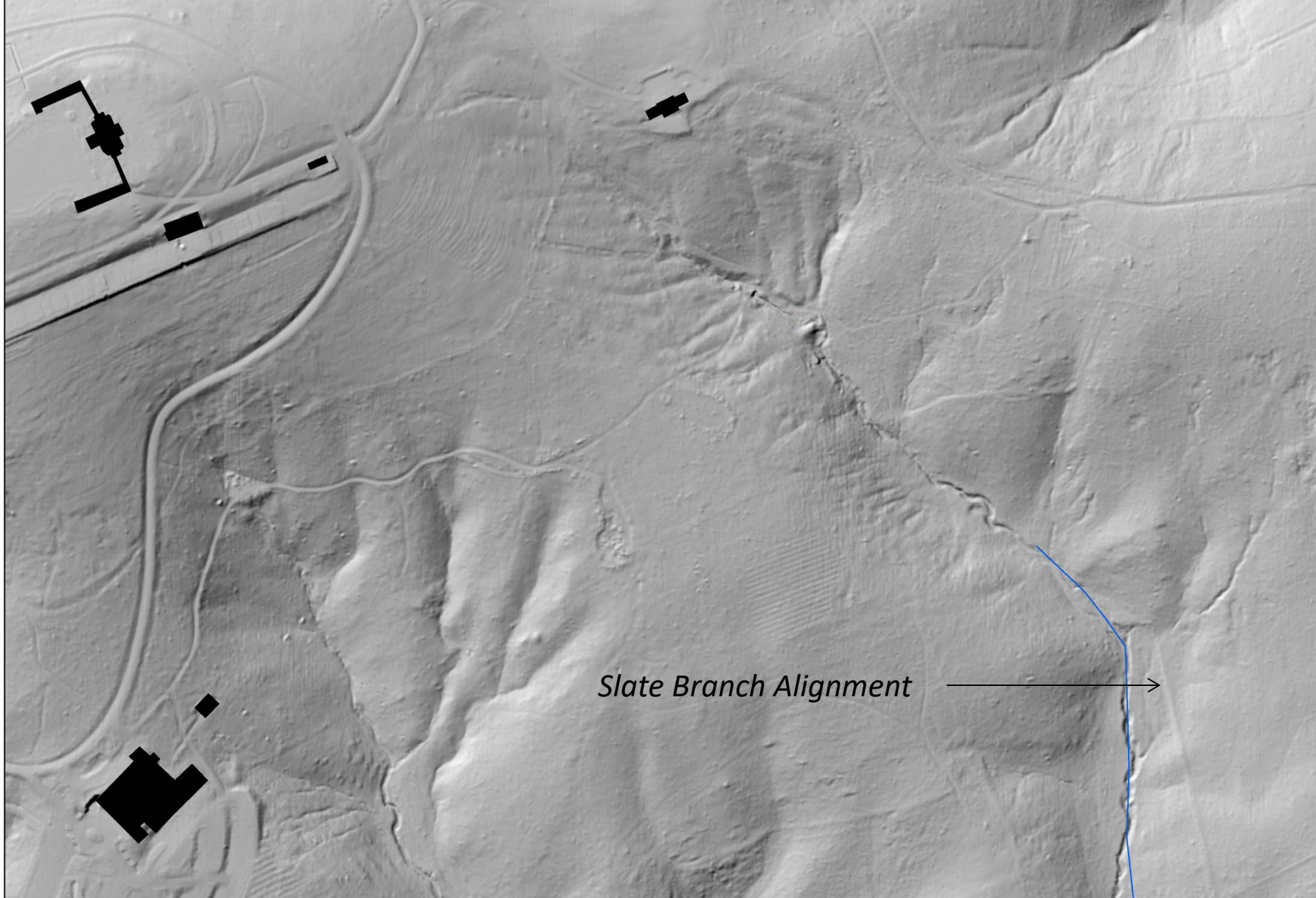




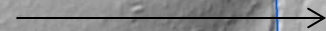
Paw Paw Valley, Units 1 & 2, West Profile







Slate Branch Alignment



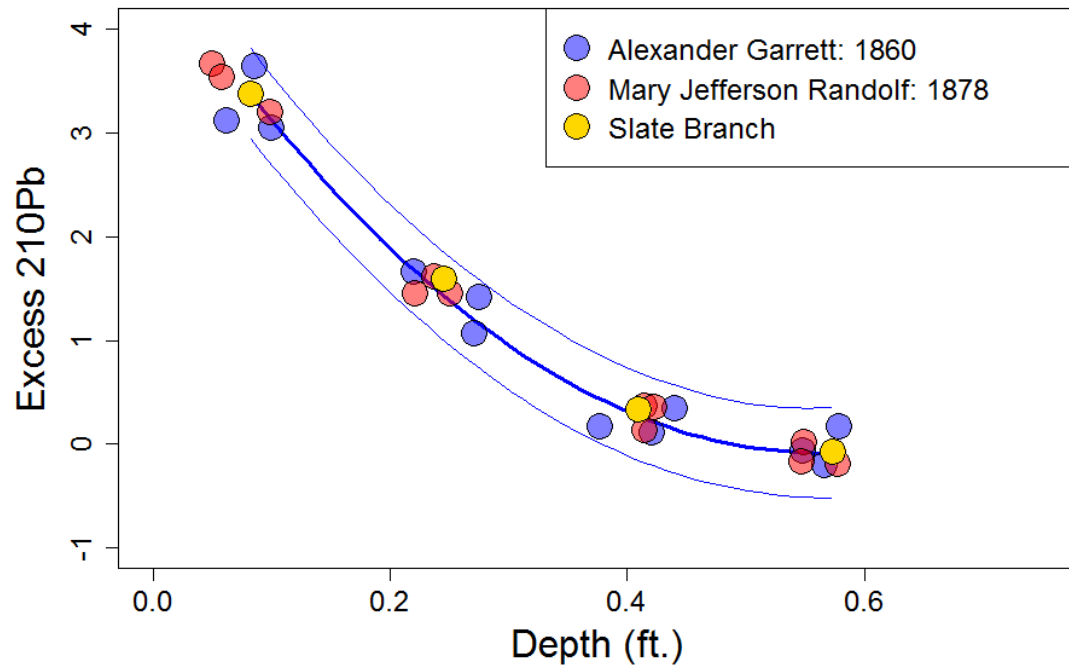
0 0.05 0.1 0.2 0.3 0.4 0.5 Kilometers

0 0.05 0.1 0.2 0.3 0.4 0.5 Miles



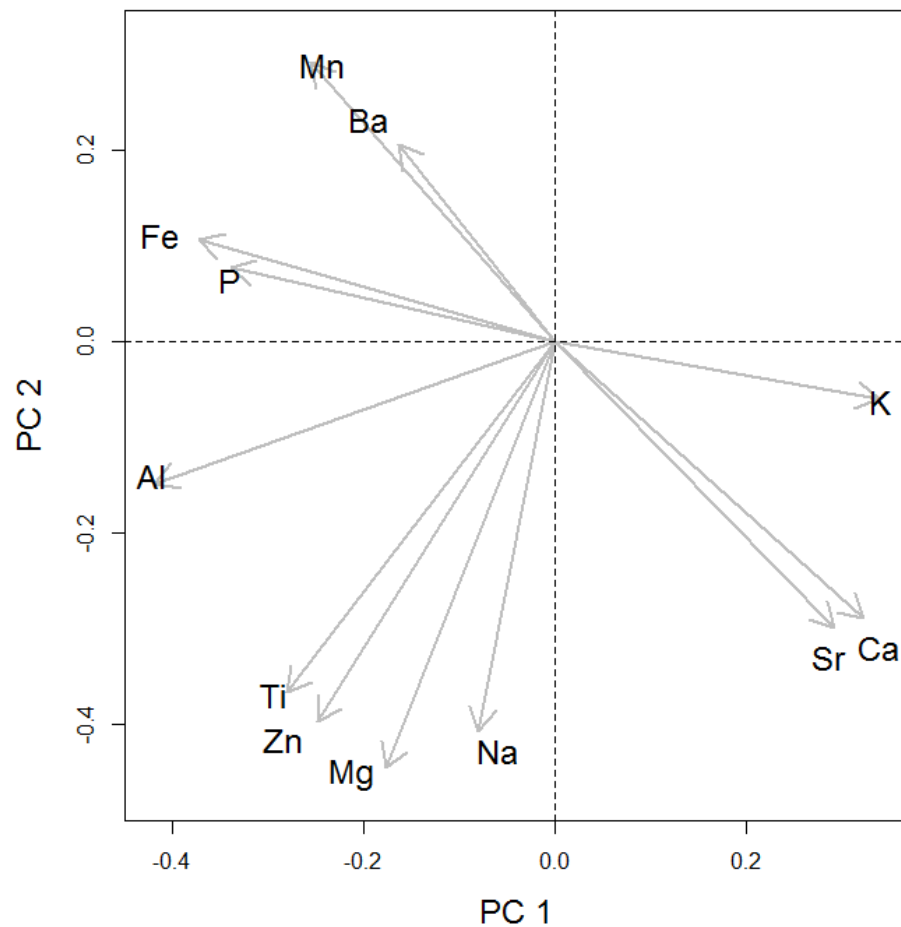
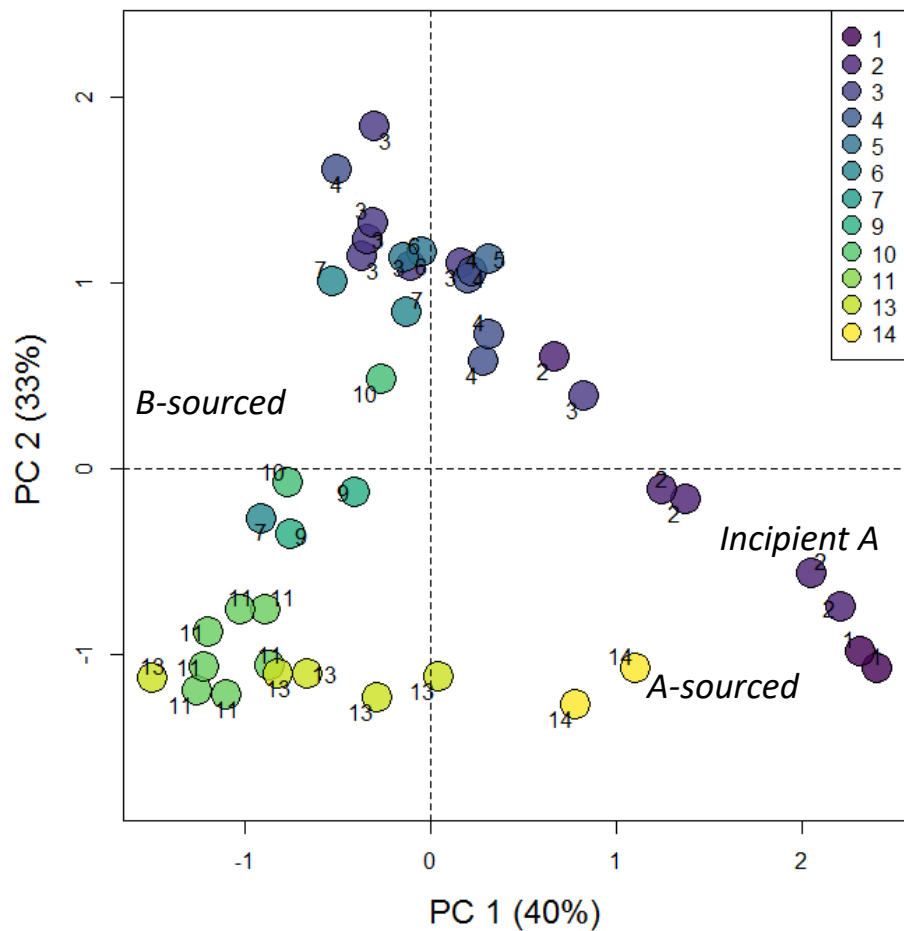
Slate Branch: Alignment: Unit 2

Dating: ^{210}Pb



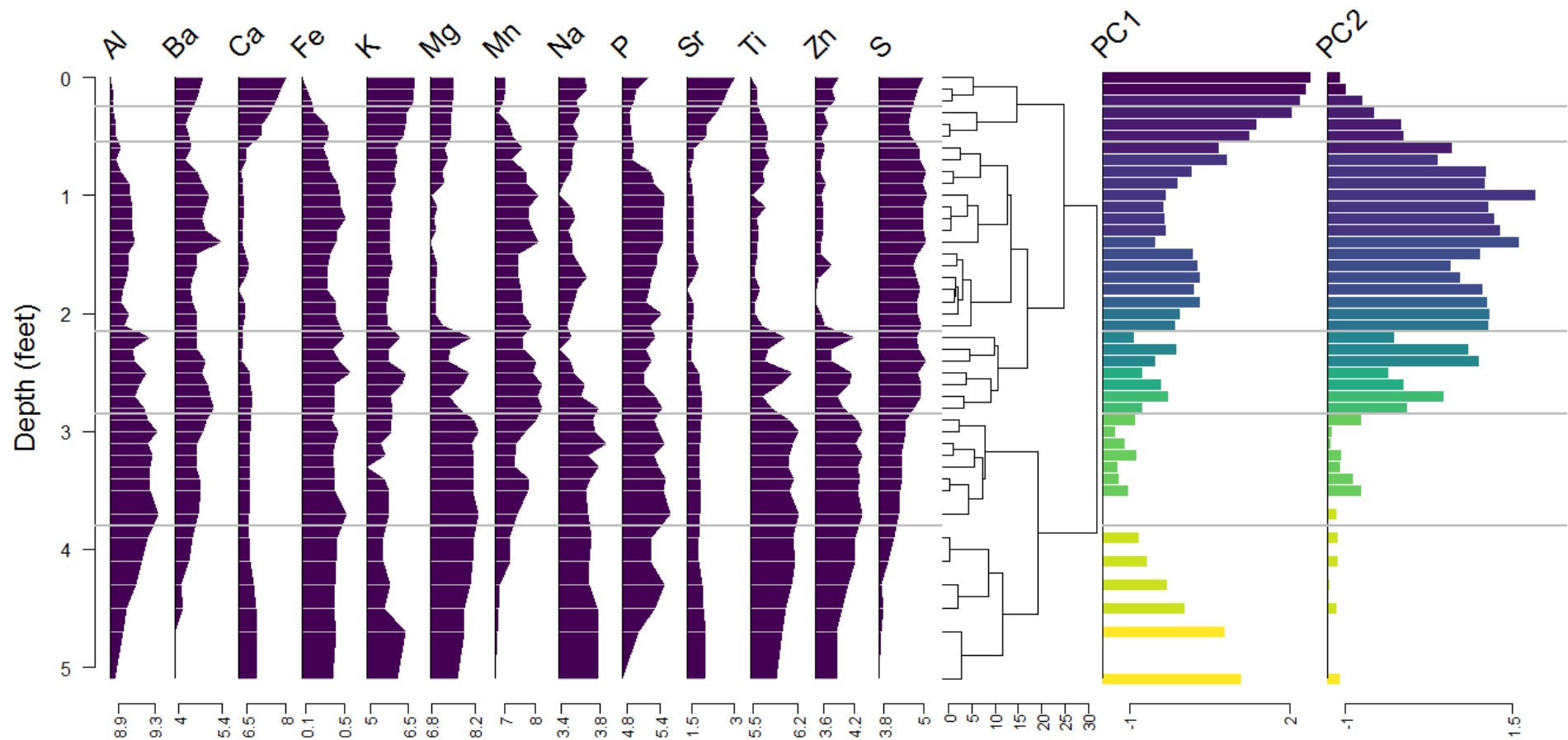
Slate Branch: Unit 2

Sediment Chemistry

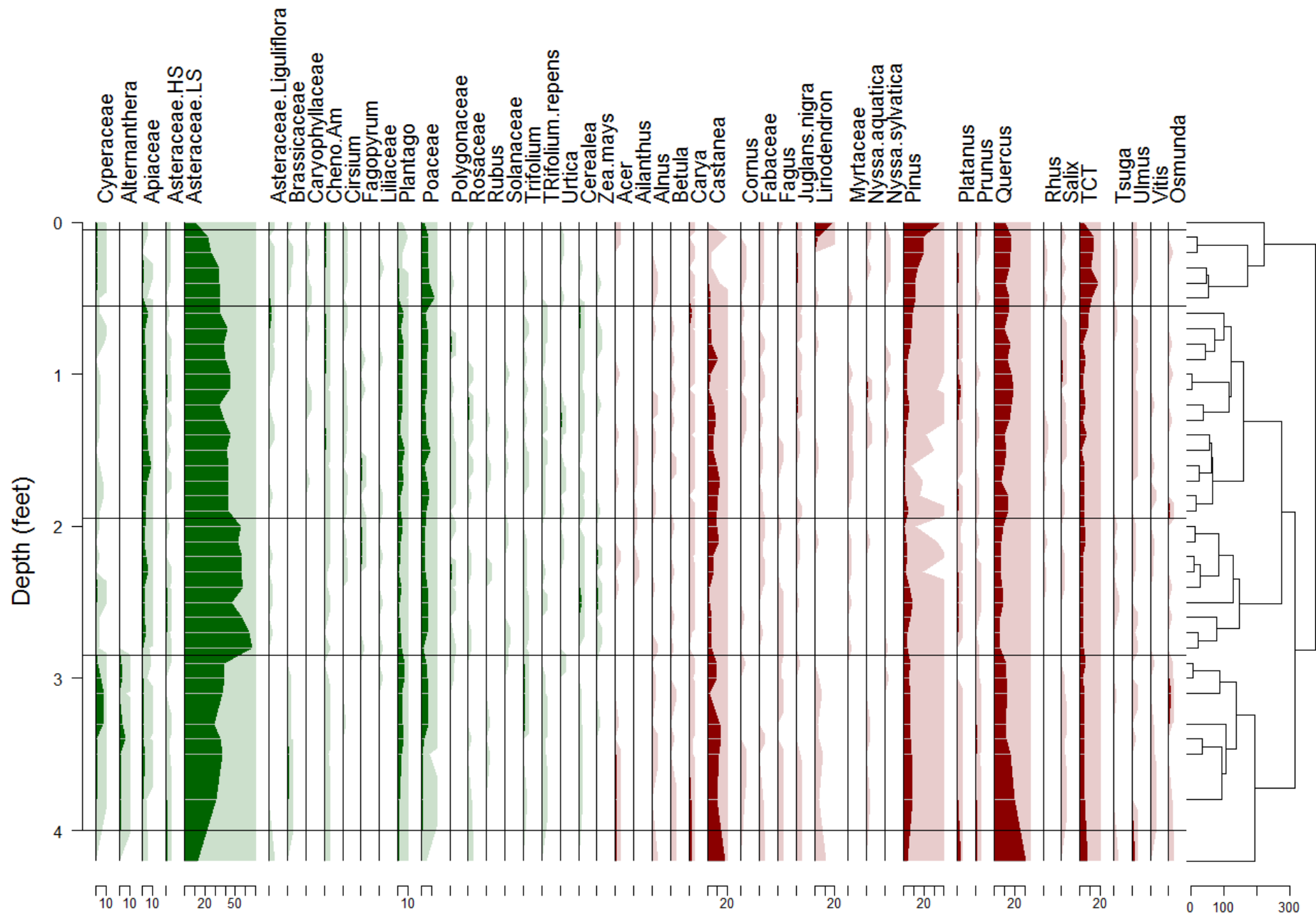


Slate Branch: Unit 2

Sediment Chemistry

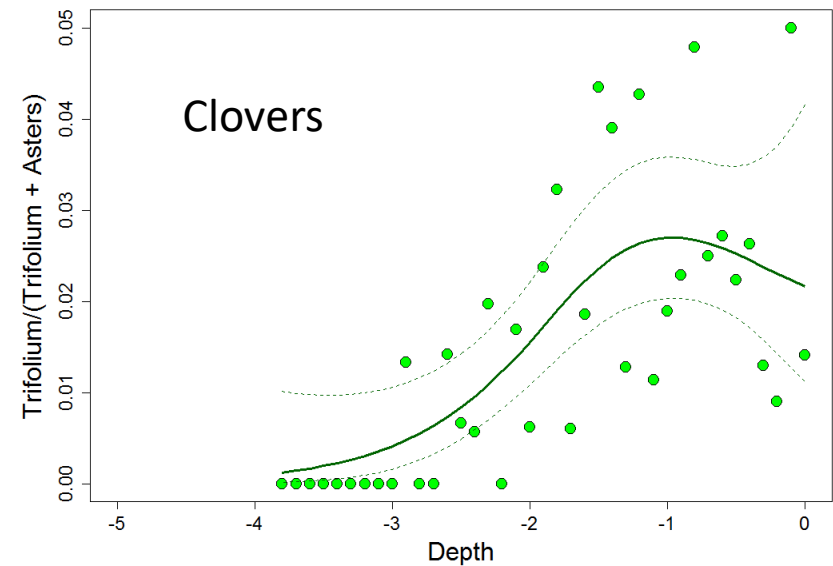
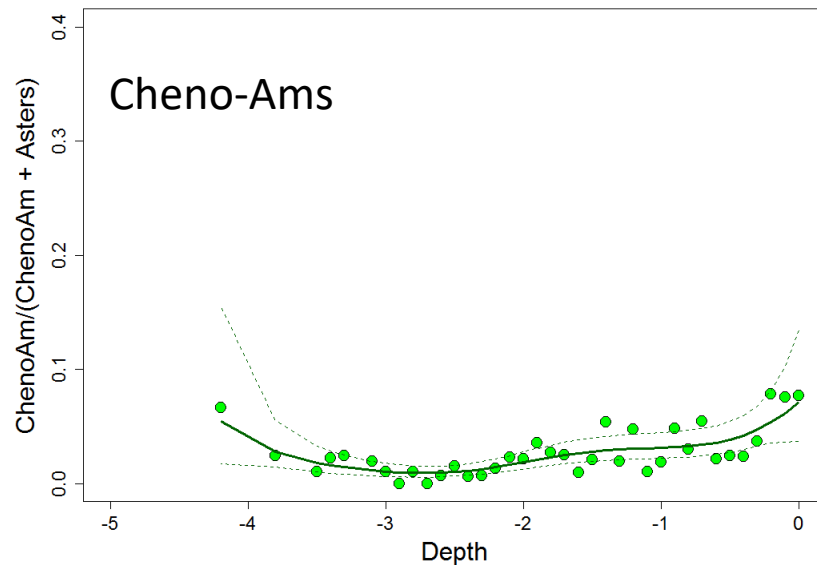
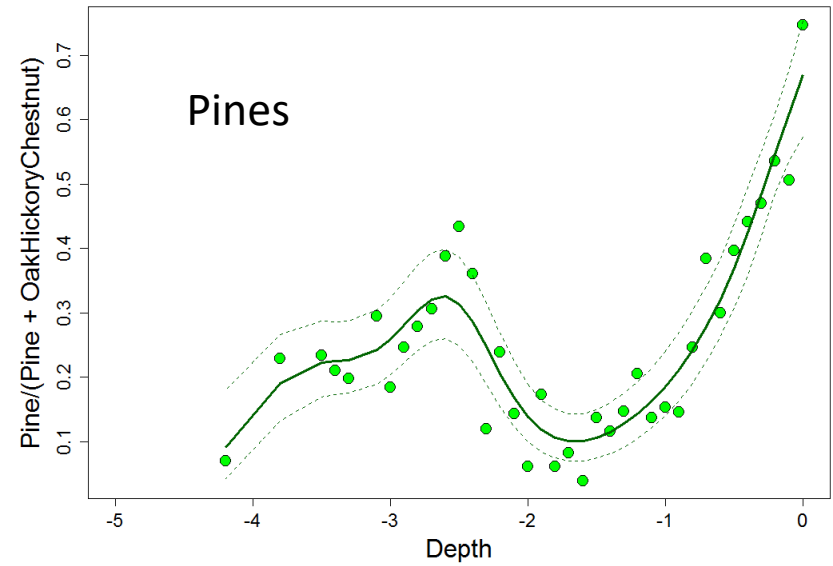
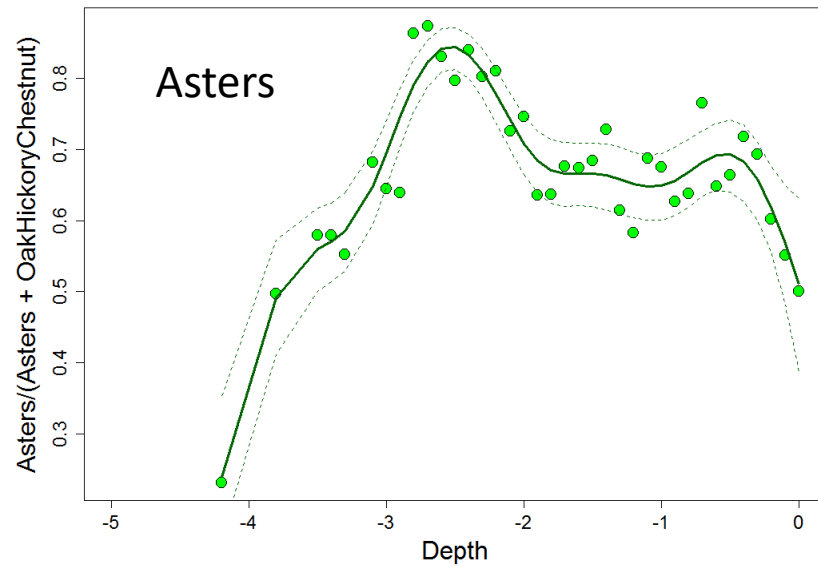


Pollen



Slate Branch: Unit 2

GAMs for Select Pollen Taxa: Logistic link, quasi-binomial errors

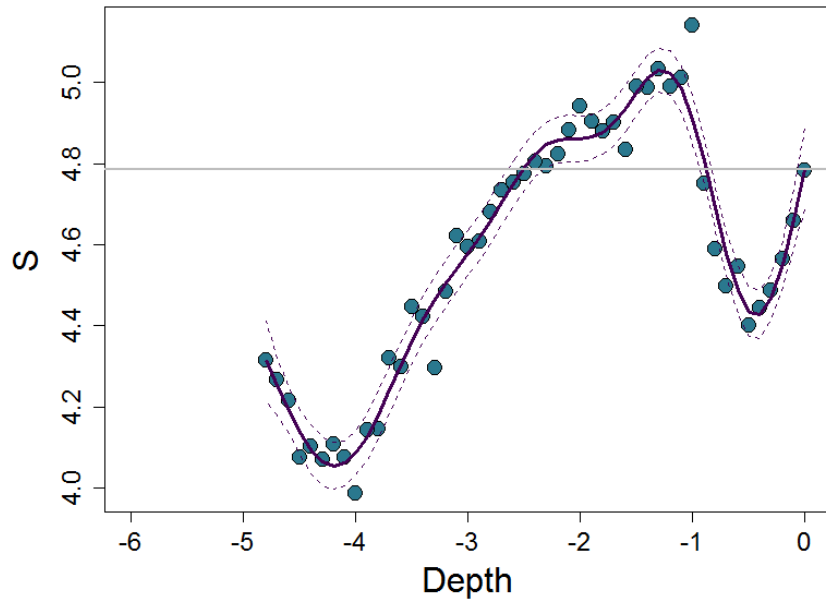


Fertilizer: Gypsum

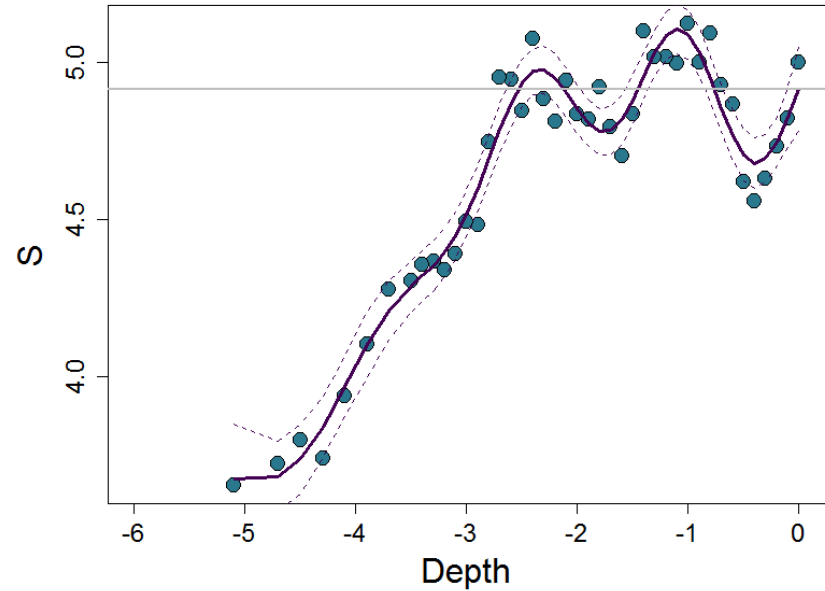
GAMs for Sulfur: identity link, Gaussian errors

Gypsum (?)

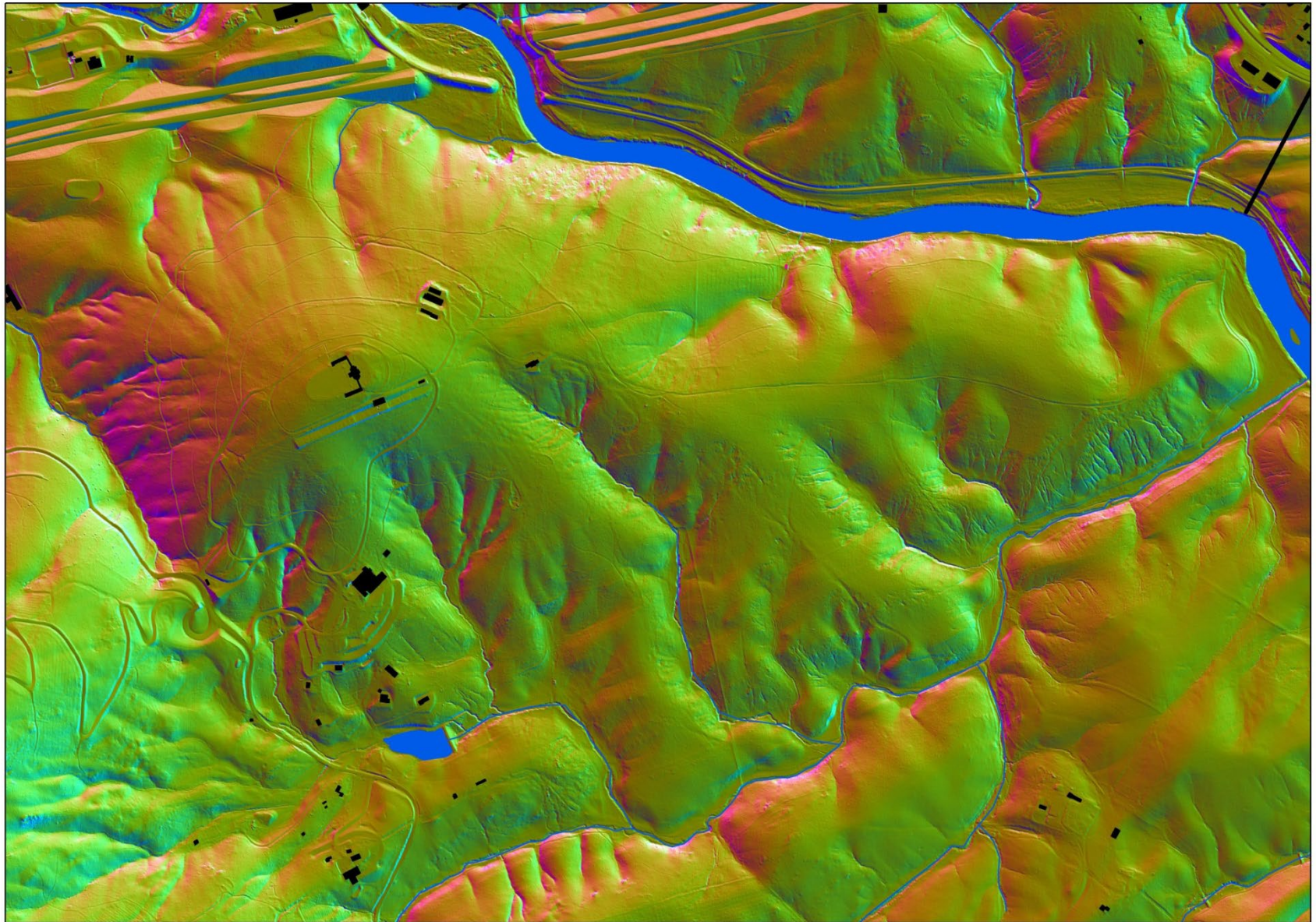
Paw Paw, Unit 1



Slate Branch, Unit 2



Gullies





Tobacco

- Hoes
- Stumps
- Field rotation
- Long fallows for sustainability
- Locally evolved

Wheat

- Plows
- No stumps!
- Permanent fields
- Crop rotations and fertilizer
- European handbooks



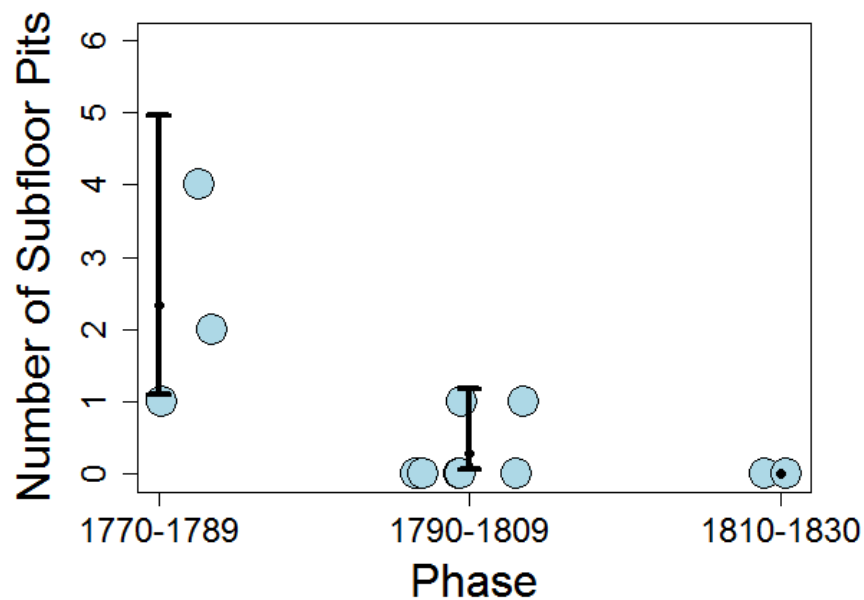
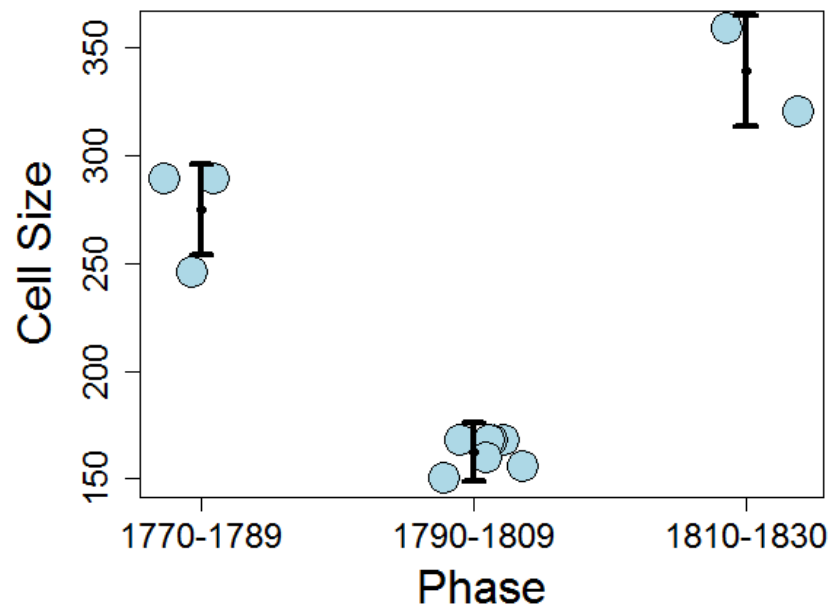
Implications

- Evidence for erosion during the tobacco period.
- But alignments built **after** the transition
- Fallow swiddens initially retained across the transition: wheat grown in newly cleared fields
- Physical traces of different kinds of labor!
- Hoe gangs vs. task groups.



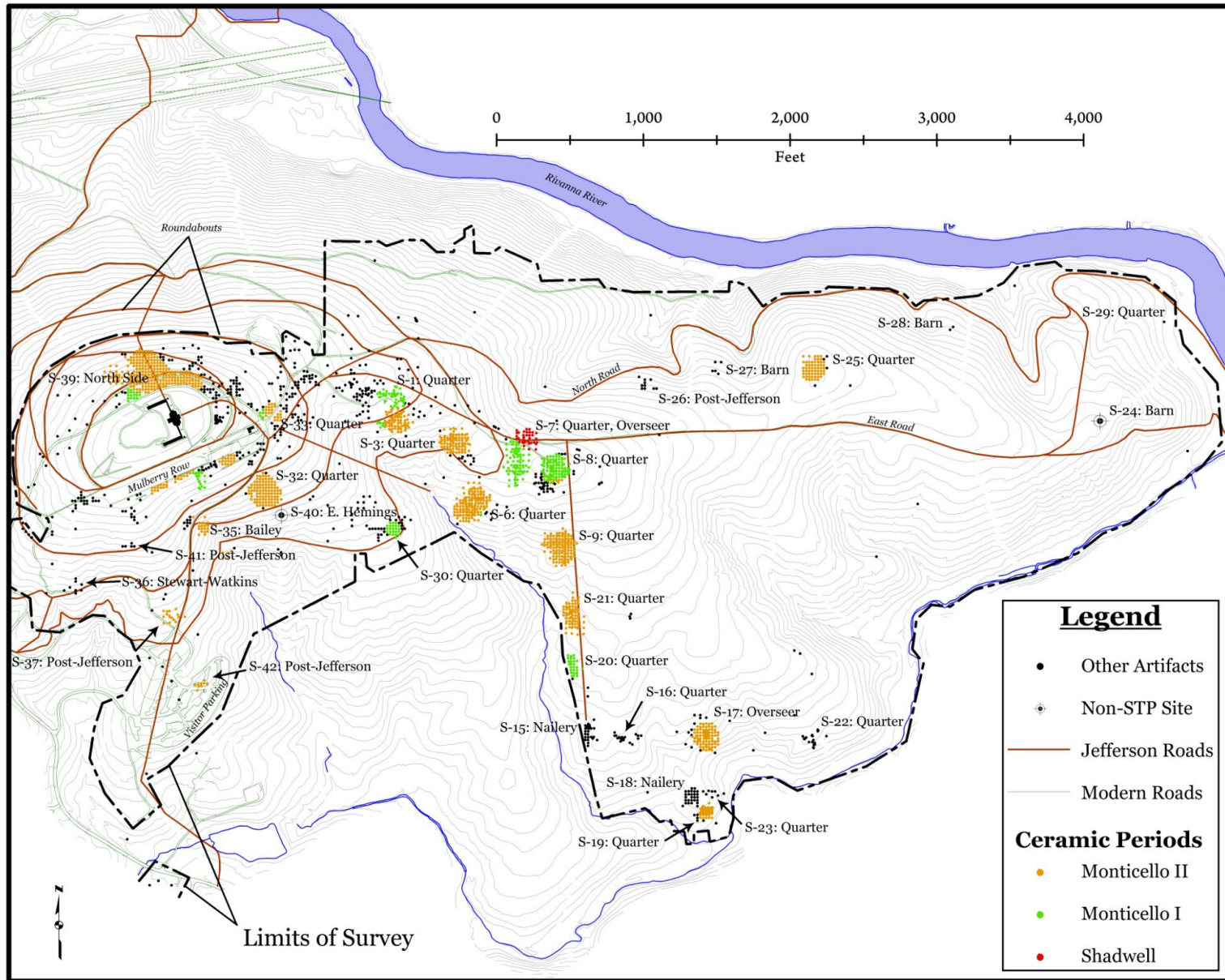
Slave Houses on Mulberry Row

Size vs. Number of Subfloor Pits

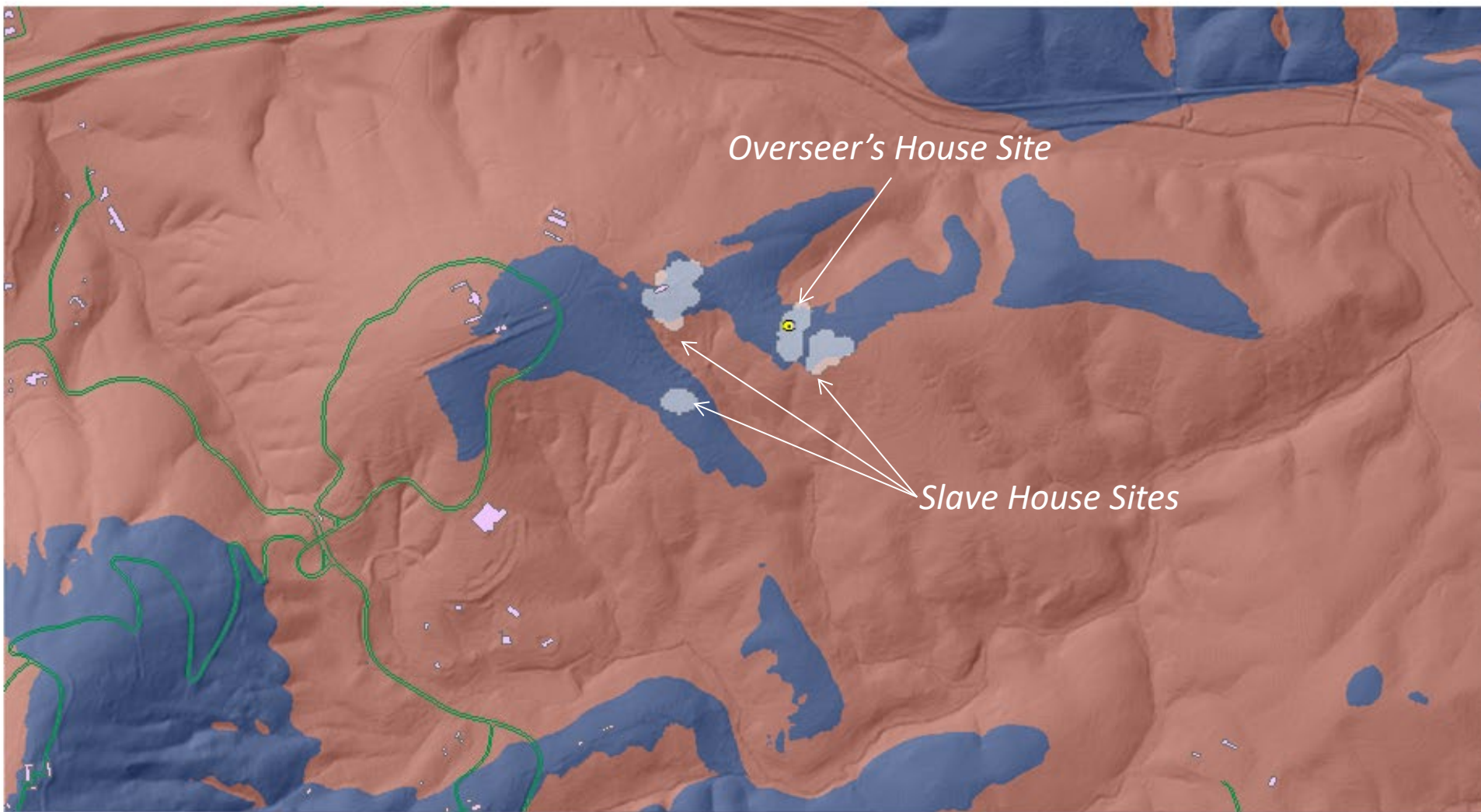


Monticello Plantation Archaeological Survey

Domestic Site Locations



Viewshed from Tobacco-Period Overseer's House



0 500 1,000 2,000 3,000 4,000
Feet



Viewshed from Wheat-Period Overseer's House

